



ADITYA UNIVERSITY

Aditya Nagar, ADB Road, Surampalem - 533 437



A D I T Y A
U N I V E R S I T Y

DATA MINING
(241AI003)

ACADAMIC REGULATION: B. Tech – 2024

LABORATORY MANUAL

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING



Vision

Aditya University aspires to be a globally recognized academic institution dedicated to quality education, cutting-edge research, and technological service to our country, and envisions itself as a beacon of holistic advancement and long-term impact, remaining dynamic in the ever-changing worlds of society, ecology, and economics.

Mission

- Aditya University pushes boundaries to design high-quality curricula and to provide students with a vibrant and relevant education that prepares them for a changing world. Our industry insights and creative teaching methods attempt to equip our students to be lifelong learners.
- Aditya University's learning environment encourages intellectual curiosity, critical thinking, and cooperation, with the goal of providing students with an immersive education that fosters creativity and innovation. Our cutting-edge facilities, interactive classrooms, and supportive faculty aim to motivate students to realize their full potential and contribute to society.
- Aditya University promotes cross-disciplinary inquiry and discovery and leads cutting-edge research and innovation. Through strategic partnerships, research grants, and a dedicated faculty, we aim to advance science, technology, and social sciences and empower students and faculty to conduct transformative research that solves real-world problems and elevates our institution globally.
- Aditya University is committed to producing world-changing business leaders and entrepreneurs through its emphasis on entrepreneurship, mentoring, and business incubation programmes.



Program Outcomes (POs) - B. Tech

After successful completion of the program, the graduates will be able to:

- | | |
|-------------|---|
| PO1 | Engineering Knowledge: Apply knowledge of mathematics, statistics, computing fundamentals and data mining concepts such as data preprocessing, classification, clustering and association rule mining to develop solutions for complex data analysis problems. (P1 to P4) |
| PO2 | Problem Analysis: Identify, formulate and analyze data related problems using appropriate data mining techniques and statistical methods to derive meaningful patterns and conclusions with consideration for practical applications. (P1 to P4) |
| PO3 | Design/Development of Solutions: Design and develop appropriate data mining models such as classification models, clustering techniques and association rule methods to extract useful knowledge from large datasets. (P5) |
| PO4 | Conduct Investigations of Complex Problems: Conduct investigations on complex datasets using data mining algorithms, perform experiments, analyze results and interpret the discovered knowledge to obtain valid conclusions. (P8) |
| PO5 | Engineering Tool Usage: Create, select and apply modern data mining and analytical tools such as WEKA or similar platforms for preprocessing, classification, clustering and association rule mining while recognizing their limitations. (P2 and P6) |
| PO6 | The Engineer and The World: Analyze and evaluate the impact of data-driven solutions in society including ethical use of data, privacy considerations and sustainable use of data technologies. (P1, P5 and P7) |
| PO7 | Ethics: Apply ethical principles while handling data, ensure responsible use of information and maintain integrity, fairness and transparency in data analysis and knowledge discovery. (P9) |
| PO8 | Individual and Collaborative Team Work: Function effectively as an individual and as a member or leader in multidisciplinary teams while performing data mining experiments, analysis and reporting |
| PO9 | Communication: Communicate effectively within the technical community by preparing reports, presenting analytical results and explaining data mining models and outcomes clearly. |
| PO10 | Project Management and Finance: Apply basic project management principles while planning, organizing and executing data mining projects involving data collection, preprocessing, modelling and evaluation. |
| PO11 | Life-Long Learning: Recognize the need for continuous learning and adaptability to emerging data mining techniques, tools and technologies and apply critical thinking to evolving data science challenges. (P8 and P10) |



Knowledge and Attitude Profile (WK)

P No.	Knowledge and Attitude Profile Description
P1	A systematic, theory-based understanding of natural sciences applicable to the discipline and awareness of relevant social sciences.
P2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
P3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
P4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
P5	Knowledge including efficient resource use, environmental impacts, whole-life cost, reuse of resources, net zero carbon and similar concepts that supports engineering design and operations in a practice area.
P6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
P7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline such as the professional responsibility of an engineer to public safety and sustainable development.
P8	Engagement with selected knowledge in the current research literature of the discipline and awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
P9	Ethics, inclusive behavior and conduct including knowledge of professional ethics, responsibilities and norms of engineering practice and awareness of the need for diversity with mutual understanding and respect.
P10	Ability to adapt to new and emerging technologies and demonstrate independent and life-long learning with critical thinking in the context of technological change.

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ELECTRICAL NETWORK ANALYSIS-II LAB

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Practice No.: 1

Illustrate KDD process and explore the Data Mining tool WEKA/KNIME for Data Preprocessing, Classification and Clustering.

Aim: To study and illustrate the Knowledge Discovery in Databases (KDD) process and to explore the data mining tool WEKA for performing data preprocessing, classification, and clustering.

Description:

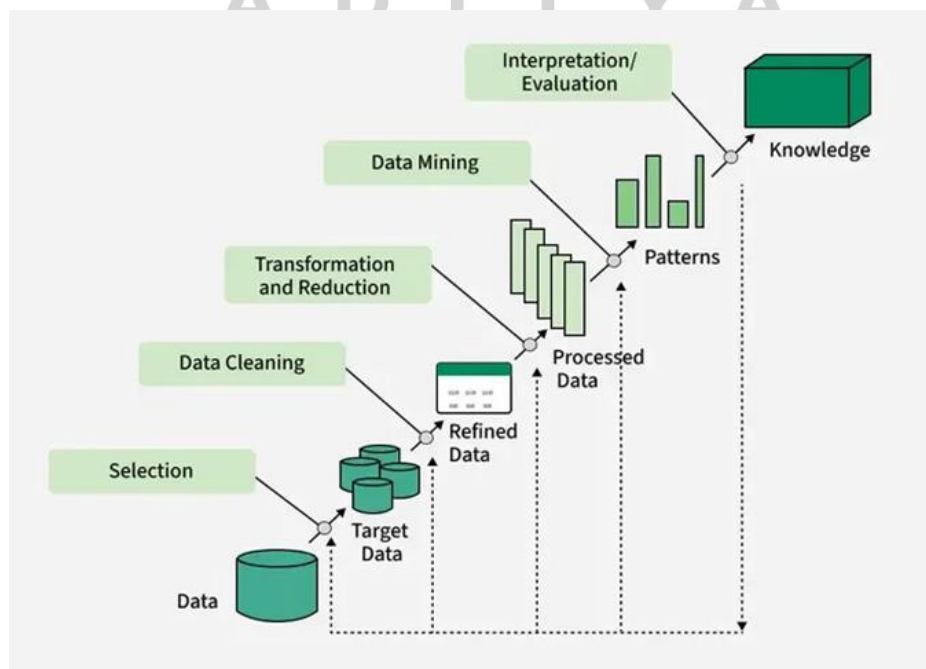
This experiment focuses on understanding the Knowledge Discovery in Databases (KDD) process, which is used to extract meaningful knowledge from large datasets. The KDD process includes stages such as data selection, preprocessing, transformation, data mining, pattern evaluation, and knowledge presentation.

In this experiment, the WEKA data mining tool is used to practically implement key data mining tasks. WEKA provides various algorithms and filters that help in data preprocessing, classification, and clustering of datasets. By using WEKA, users can easily analyze data, apply machine learning techniques, and interpret the results effectively.

Knowledge Discovery in Databases (KDD) Process

Introduction

Knowledge Discovery in Databases (KDD) is the systematic process of extracting meaningful knowledge and patterns from large volumes of data. It involves multiple steps, of which Data Mining is the central stage.



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Stages of the KDD Process :

1. Data Selection

- Selection of relevant data from databases, data warehouses, or multiple data sources.
 - Irrelevant and redundant data are excluded.
-

2. Data Preprocessing (Data Cleaning)

- Improves data quality by removing errors and inconsistencies.

Activities include:

- Handling missing values
 - Removing duplicate records
 - Noise reduction
 - Outlier detection
-

3. Data Transformation

- Converts data into a suitable format for mining.

Techniques:

- Normalization
 - Discretization
 - Aggregation
 - Feature selection
-

4. Data Mining

- Application of algorithms to extract patterns from data.

Common tasks:

- Classification
- Clustering
- Association rule mining
- Regression

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5. Pattern Evaluation

- Identifies useful and interesting patterns.
- Filters out irrelevant or redundant information.

6. Knowledge Presentation

- Presents discovered knowledge using graphs, reports, and visualizations for easy understanding.

Conclusion:

Thus, the Knowledge Discovery in Databases (KDD) process provides a systematic and step-by-step approach to extract useful, valid, and meaningful knowledge from large datasets. Each stage of KDD plays an important role in improving data quality and ensuring accurate and reliable results.

Data Mining Tool: WEKA

Introduction to WEKA

WEKA (Waikato Environment for Knowledge Analysis) is an open-source data mining tool developed by the University of Waikato, New Zealand. It is written in Java and provides a collection of machine learning algorithms for data analysis.

Key Features of WEKA Simple graphical user interface

- Supports data preprocessing, classification, and clustering.
- Accepts ARFF and CSV file formats.
- Suitable for academic and small-scale applications.



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Installation Steps :

Step 1: Open WEKA Official Website

- Open any web browser (Chrome / Edge / Firefox).
- Go to the official WEKA website.
- On the homepage, look for the Download option.
- Click on WEKA 3.8.6 (as shown in your photo).

Step 2: Select Operating System

After clicking download, you will see options for different operating systems:

- Windows
- macOS
- Linux

Click on Windows.

Step 3: Download the Installer File

- The file name will look like: weka-3-8-6.exe
- Click on the file.
- The download will start.
- Wait until it reaches 100% completed.
- Open the Downloaded File Go to the Downloads folder

Step 4: WEKA Setup Wizard – Welcome Screen



- The WEKA Setup window opens.

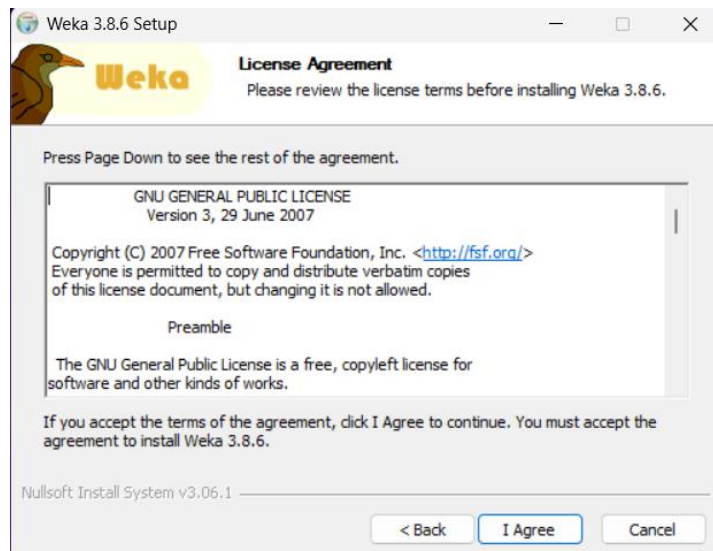
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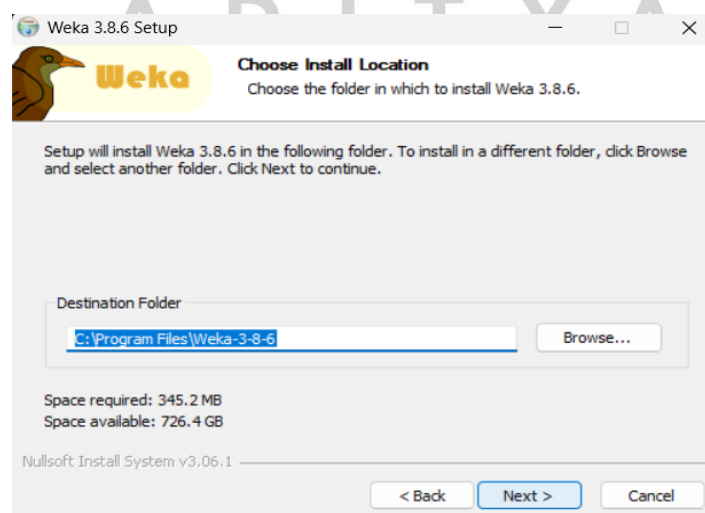
- You will see a Welcome to WEKA Setup message.
- Click Next.

Step 5: License Agreement



- The license agreement window appears.
- Select I Agree or Accept.
- Click Next.

Step 6: Choose Installation Location



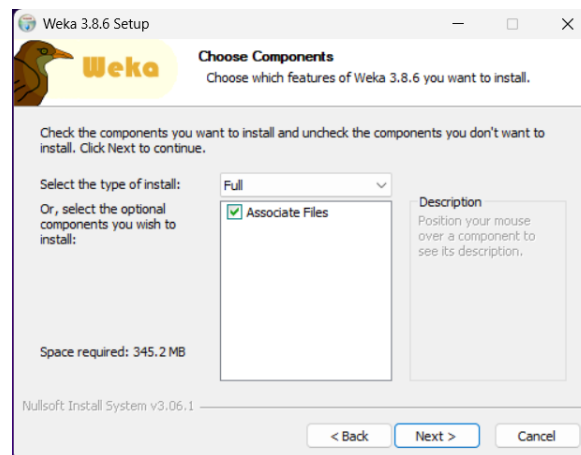
- Default path is usually:
C:\Program Files\Weka-3-8
- Keep it unchanged.
- Click Next.

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Step 7: Select Components

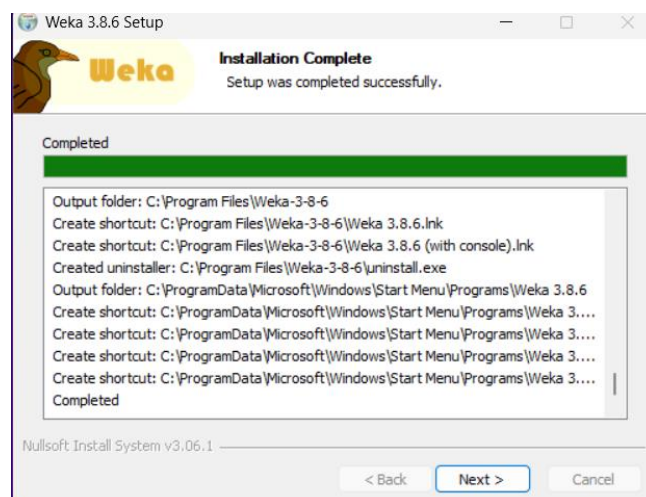


- Default components will be selected automatically.
- Do not change anything.
- Click Next.

Step 8: Install WEKA

- Click Install.
- The installation progress bar will move.
- Wait until installation completes.

Step 9: Finish Installation



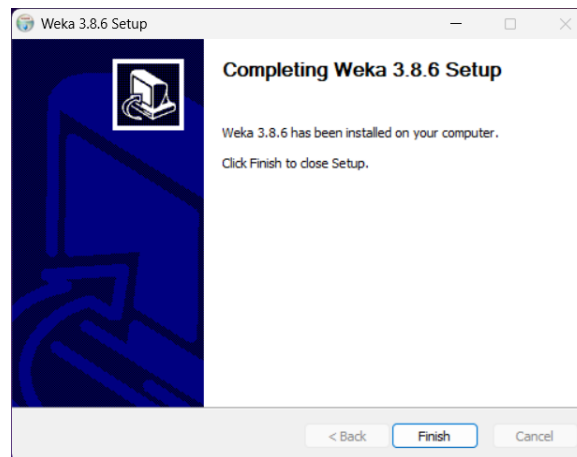
- After completion, click Finish.
- WEKA installation is now complete.

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Installation Successful



You have successfully installed WEKA 3.8.6 for Data Mining tasks.

WEKA for Data Preprocessing:

Handling Missing Values

WEKA provides filters to manage missing data by replacing them with mean, median, mode, or user-defined values. This improves data quality and helps in accurate analysis.

Data Normalization and Discretization

Normalization scales numeric data into a specific range, while discretization converts continuous attributes into categorical intervals. These processes make data suitable for mining algorithms.

Attribute Selection Using Filters

WEKA allows selection of relevant attributes by removing redundant or irrelevant features using filters. This reduces complexity and improves model performance.

WEKA for Classification:

J48 (Decision Tree)

J48 is an implementation of the C4.5 algorithm. It creates a tree structure for decision-making and is easy to interpret.

Naive Bayes

Naive Bayes is a probabilistic classifier based on Bayes' theorem. It is fast and effective for large datasets.

Random Forest

Random Forest is an ensemble method that builds multiple decision trees and combines their results to improve accuracy.

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Support Vector Machine (SMO)

SMO is WEKA's implementation of SVM. It separates data into classes using optimal hyperplanes.

Accuracy and Confusion Matrix

WEKA evaluates classification performance using accuracy measures and confusion matrices to show correct and incorrect predictions

WEKA for Clustering:

K-Means

K-Means groups data into a predefined number of clusters based on similarity.

Hierarchical Clustering

This method creates a tree-like structure of clusters, showing relationships between data points.

EM Algorithm

Expectation Maximization (EM) forms clusters based on probability distribution models.

2D Cluster Visualization

WEKA provides 2D plots to visually represent clusters, making pattern interpretation easier.

Conclusion :

WEKA is a powerful and user-friendly data mining tool that simplifies the process of analyzing datasets. It effectively supports data preprocessing, classification, and clustering, allowing users to handle missing values, select important attributes, apply machine learning algorithms, and visualize results. By using WEKA, meaningful patterns and knowledge can be extracted from data efficiently, making it an essential tool for academic, research, and practical data analysis applications.

CO Mapped: CO1

POs Mapped: PO1, PO2, PO3, PO5, PO8, PO9, PO11

PSOs Mapped: PSO2

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Practice No.: 2

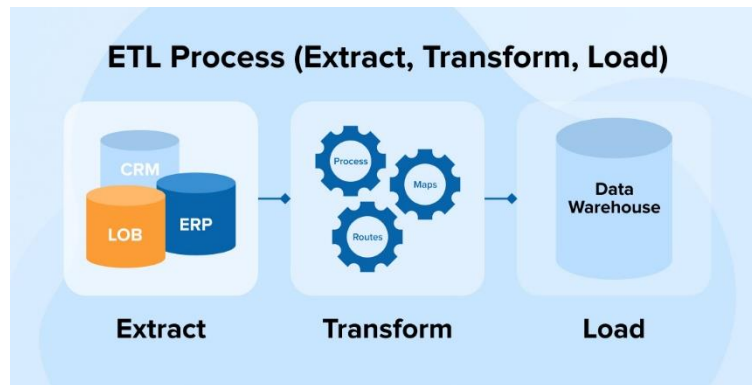
Explain ETL process important for moving data in the data warehouse.

Aim: To study and explain the ETL (Extract, Transform, Load) process and understand its importance in moving and preparing data for a Data Warehouse.

Introduction

In a data warehousing environment, data is collected from multiple heterogeneous sources such as databases, files, and applications. Since this data originates from different systems, it is often inconsistent, incomplete, and stored in various formats. Before this data can be used effectively for analysis and decision-making, it must be properly processed, cleaned, and standardized prior to being stored in the data warehouse.

What is ETL?



ETL stands for Extract, Transform, and Load. It is a systematic process used in data warehousing to extract data from various source systems, transform the extracted data into a consistent and suitable format, and finally load the processed data into a data warehouse for analysis and decision-making purposes.

Stages of ETL Process

1. Extract

Extract Stage is the first step in ETL where data is read from multiple source systems such as:

- Databases (using SQL/JDBC connections)
- Flat files like CSV/TXT (read_csv())
- Excel files (read_excel())
- Applications or web systems via APIs
- Transactional/operational systems

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The data pulled from all these locations is stored in a staging area (DataFrame or temp DB) for further processing.

Types implemented in code:

- **Full Extraction:** scripts that read the entire file/table directly with `pd.read_csv()` or `pd.read_sql()`.
- **Incremental Extraction:** code that filters only new records by comparing IDs between old and new datasets.

Purpose achieved by the code:

- Gathers heterogeneous raw data into one structure.
- Prepares a common staging dataset for the Transform and Load stages.

2. Transform

The transformation codes implement key activities such as `drop_duplicates()` to remove duplicate records, `fillna()` to handle missing/null values, and string functions to correct inconsistent text data.

They use `astype()` and `to_datetime()` for data type conversion and formatting logic to standardize dates and numeric fields.

Business filters in the code represent application of business rules (e.g., keeping only completed transactions).

`groupby().sum()/mean()/count()` performs aggregation, matching analytical preparation.

Purpose achieved by the code:

- Improves overall data quality,
- Ensures consistency and accuracy,
- Produces a clean dataset suitable for data-warehouse analytics and BI reporting.

3. Load

Loading is the final stage of the ETL process. In this stage, the transformed data is stored in the data warehouse.

Types of Loading:

- **Full Load:** Entire transformed data is loaded.
- **Incremental Load:** Only new or updated data is loaded.

Data is Loaded Into:

- Fact tables

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- Dimension tables

Purpose of Loading:

- To make data available for reporting, analysis, and decision-making.

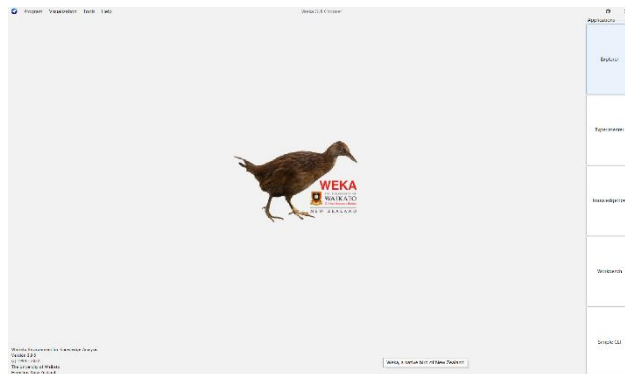
Importance of ETL in Data Warehouse

- Integrates data from multiple heterogeneous sources
- Improves data accuracy and consistency
- Enhances data quality
- Reduces redundancy and errors
- Enables efficient querying and reporting
- Supports business intelligence and data mining

Procedure:

Step 1: Install and Open Weka

- Download and install the Weka tool on your system.
- Open the application.
- Click on Explorer from the Weka GUI.



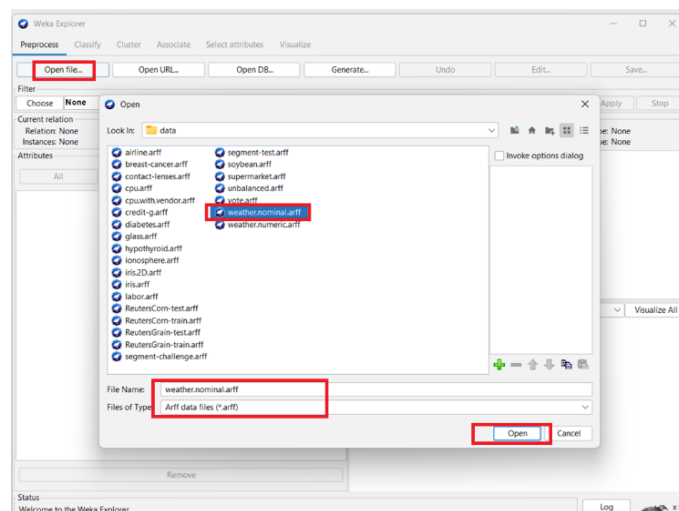
Step 2: Extract Data (Loading Dataset)

- In the Explorer window, select the Preprocess tab.
- Click Open File.
- Choose the dataset file (CSV or ARFF format).
- The dataset will be loaded and displayed with attributes and instances.
- This step represents the Extract phase, where data is taken from a source file.

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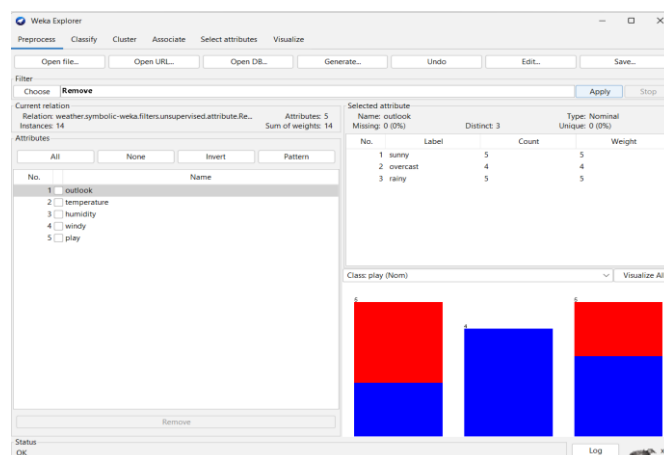
Step 3: Transform Data (Preprocessing)

In the Preprocess tab, select Choose Filter.

Apply filters such as:

Apply Remove Filter (Remove Unnecessary Attributes)

- Click Choose.
- Select the following path
unsupervised → attribute → Remove
- Click on Remove.
- Enter the attribute index number you want to remove (example: 2).
- Click OK.
- Click Apply.
- The selected attribute will be removed from the dataset.



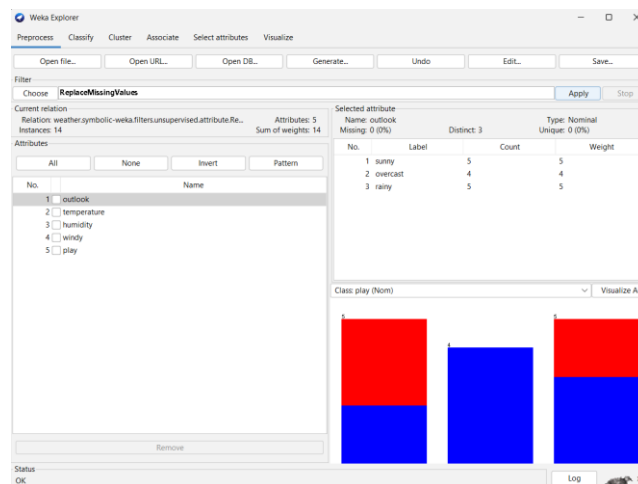
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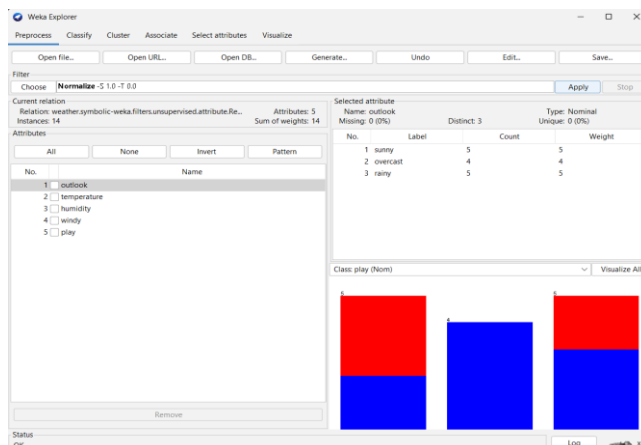
Apply ReplaceMissingValues Filter

- Click Choose again.
- Select:
unsupervised → attribute → ReplaceMissingValues
- Click ReplaceMissingValues.
- Click Apply.
- Weka will automatically replace missing values in the dataset.



Apply Normalize or Standardize Filter

- Click Choose again.
- Select:
unsupervised → attribute → Standardize / Normalize Filter
- Click the selected filter.
- Click Apply.
- This adjusts the data values into a standard range.



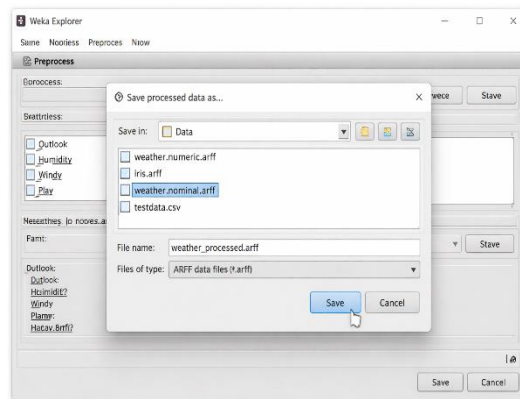
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Step 4: Load Data (Save Processed Dataset)

- After preprocessing, click Save in the Preprocess tab.
- Save the transformed dataset in ARFF or CSV format.
- The processed data can now be used for data analysis or machine learning tasks.
- This step represents the Load phase, where the cleaned data is stored for further use.



Conclusion:

Using the Weka tool, the ETL (Extract, Transform, Load) process was carried out on the dataset. The data was first extracted by importing it into Weka, then transformed using preprocessing techniques such as filtering and handling missing values, and finally loaded by saving the processed dataset. This process helps in cleaning, organizing, and preparing data for further analysis and data mining tasks, making the data more accurate and useful for decision-making.

CO Mapped: CO1

POs Mapped: PO1, PO2, PO3, PO5, PO8, PO9, PO11

PSOs Mapped: PSO2

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Practice No.:3

Apply Statistical Descriptions and Measures of Similarity and Dissimilarity on given data

Aim: To apply statistical descriptions and measures of similarity and dissimilarity on the given student academic performance data.

Tool Used:

WEKA Data Mining Tool

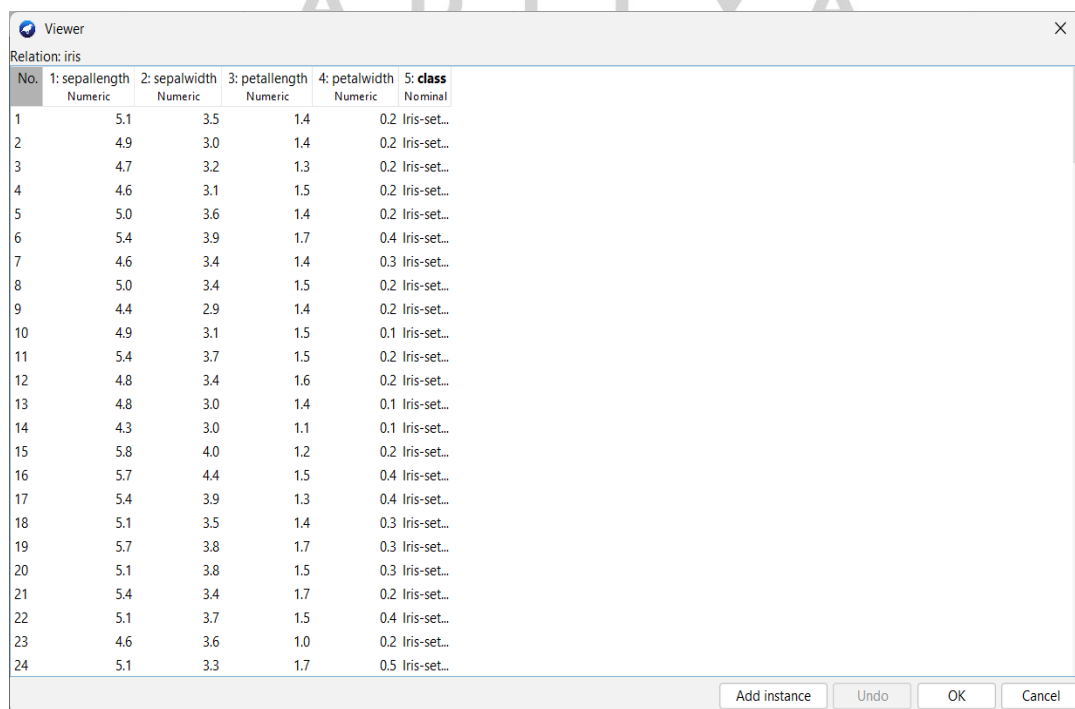
Dataset Used – Iris Dataset:

The Iris dataset is a built-in dataset available in WEKA. It contains 150 instances of iris flowers with numeric measurements of sepals and petals.

Attributes

1. Sepal Length (numeric)
2. Sepal Width (numeric)
3. Petal Length (numeric)
4. Petal Width (numeric)
5. Class (Iris-setosa, Iris-versicolor, Iris-virginica)

This dataset is suitable because all attributes are numeric, which helps in calculating statistical measures and distance-based similarity.



No.	1: sepal length Numeric	2: sepal width Numeric	3: petal length Numeric	4: petal width Numeric	5: class Nominal
1	5.1	3.5	1.4	0.2	Iris-set...
2	4.9	3.0	1.4	0.2	Iris-set...
3	4.7	3.2	1.3	0.2	Iris-set...
4	4.6	3.1	1.5	0.2	Iris-set...
5	5.0	3.6	1.4	0.2	Iris-set...
6	5.4	3.9	1.7	0.4	Iris-set...
7	4.6	3.4	1.4	0.3	Iris-set...
8	5.0	3.4	1.5	0.2	Iris-set...
9	4.4	2.9	1.4	0.2	Iris-set...
10	4.9	3.1	1.5	0.1	Iris-set...
11	5.4	3.7	1.5	0.2	Iris-set...
12	4.8	3.4	1.6	0.2	Iris-set...
13	4.8	3.0	1.4	0.1	Iris-set...
14	4.3	3.0	1.1	0.1	Iris-set...
15	5.8	4.0	1.2	0.2	Iris-set...
16	5.7	4.4	1.5	0.4	Iris-set...
17	5.4	3.9	1.3	0.4	Iris-set...
18	5.1	3.5	1.4	0.3	Iris-set...
19	5.7	3.8	1.7	0.3	Iris-set...
20	5.1	3.8	1.5	0.3	Iris-set...
21	5.4	3.4	1.7	0.2	Iris-set...
22	5.1	3.7	1.5	0.4	Iris-set...
23	4.6	3.6	1.0	0.2	Iris-set...
24	5.1	3.3	1.7	0.5	Iris-set...

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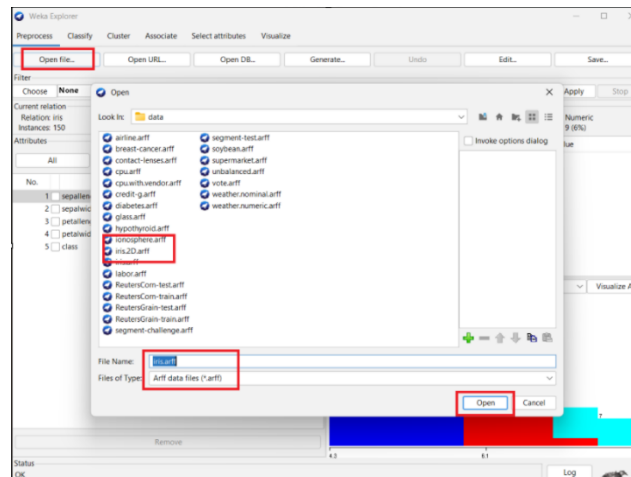
Procedure:

Step 1: Open WEKA

- Double click on WEKA Tool
- Click on Explorer

Step 2: Load the Iris Dataset

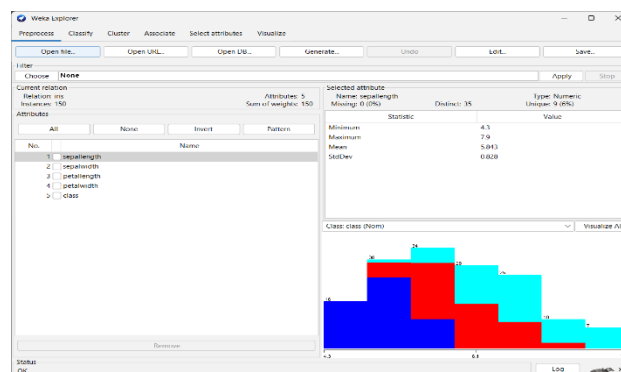
- Go to Preprocess tab
- Click Open file
- Navigate to: WEKA → data → iris.arff
- Click Open
- The dataset with 150 instances and 5 attributes will be loaded



Step 3: Observe Dataset Information

In the Preprocess tab, observe:

- Relation name: iris
- Number of instances: 150
- Number of attributes: 5



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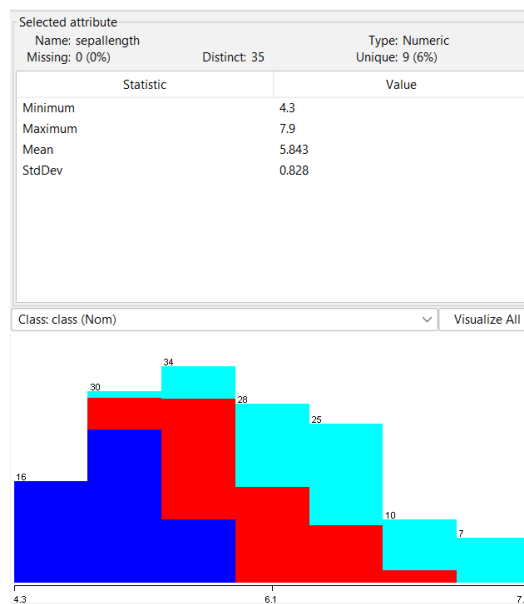
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Step 4: Apply Statistical Description

Select each numeric attribute one by one:

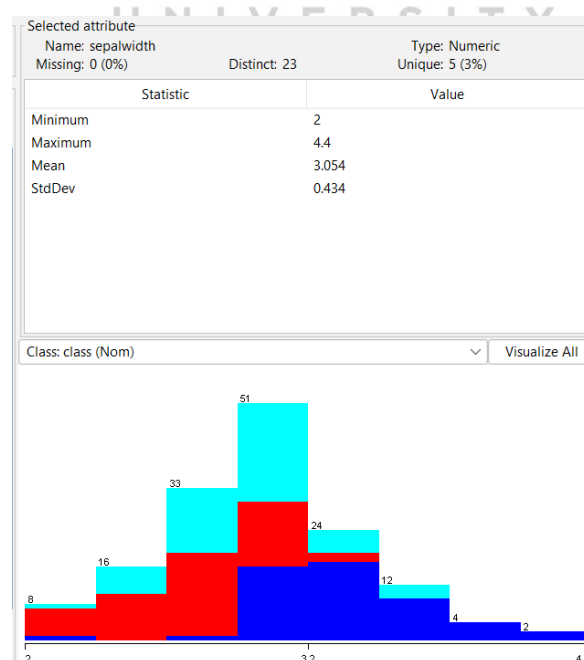
(a) Sepal Length

- Click sepal length
- Note: Minimum, Maximum, Mean, Standard Deviation



(b) Sepal Width

- Click sepal width
- Note the statistical values



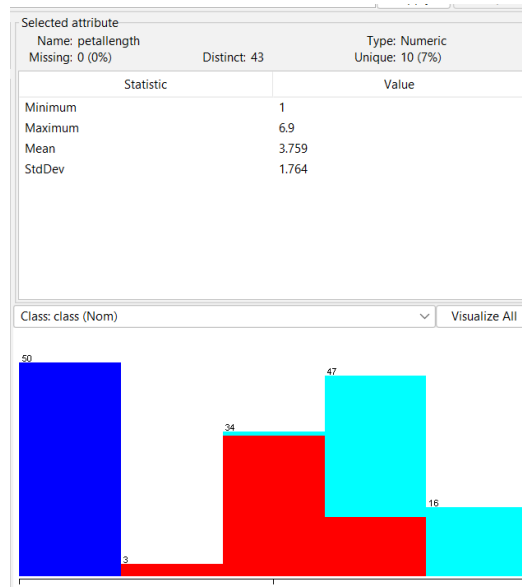
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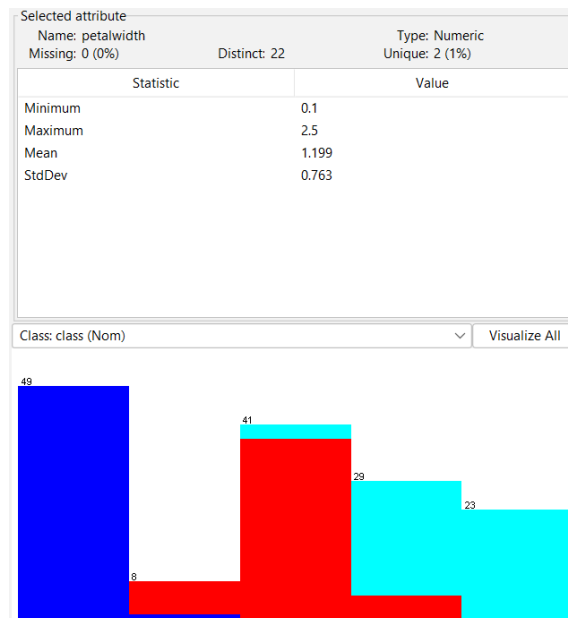
(c) Petal Length

- Click petal length
- Note the statistical values



(d) Petal Width

- Click petal width
- Note the statistical values



These values represent the statistical summary of the dataset.

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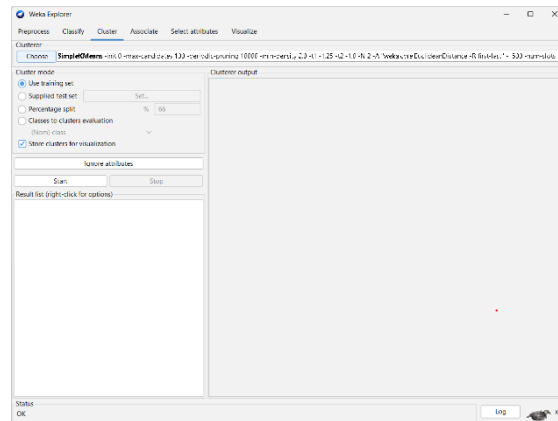
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Step 5: Move to Cluster Tab for Similarity Measure

- Click on Cluster tab

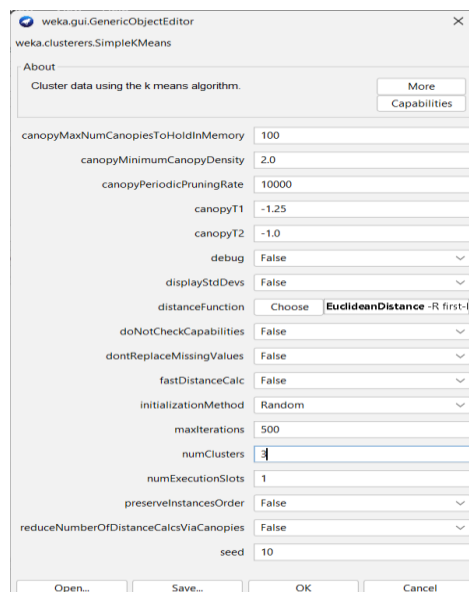
Step 6: Choose Clustering Algorithm

- Click Choose
- Select SimpleKMeans



Step 7: Configure SimpleKMeans

- Click on the text SimpleKMeans
- Set the following:
 - Number of clusters = 3
 - Distance Function = EuclideanDistance
- Click OK



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Step 8: Start Clustering

- Click Start
- WEKA will process and create clusters

Step 9: Observe Cluster Output

WEKA displays:

- Cluster centroids (mean values of attributes)
- Number of instances in each cluster

These centroids represent statistical averages of similar records.

```
Clusterer output
=== Run information ===

Scheme:      weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning
Relation:    iris
Instances:    150
Attributes:   5
              sepalwidth
              sepalwidth
              petalwidth
              petalwidth
              class
Test mode:    evaluate on training data

=== Clustering model (full training set) ===

kMeans
=====

Number of iterations: 3
Within cluster sum of squared errors: 7.817456892309574

Initial starting points (random):

Cluster 0: 6.1,2.9,4.7,1.4,Iris-versicolor
Cluster 1: 6.2,2.9,4.3,1.3,Iris-versicolor
Cluster 2: 6.9,3.1,5.1,2.3,Iris-virginica

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute      Full Data      Cluster#
              (150.0)      (50.0)      (50.0)      (50.0)
=====
sepalwidth     5.8433         5.936         5.006         6.588
sepalwidth     3.054          2.77          3.418         2.974
petalwidth     3.7587         4.26          1.464         5.552
petalwidth     1.1987         1.326         0.244         2.026
class          Iris-setosa Iris-versicolor Iris-setosa Iris-virginica

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      50 ( 33%)
1      50 ( 33%)
2      50 ( 33%)
```

Step 10: Interpret Similarity

Similarity is measured using Euclidean Distance in the SimpleKMeans algorithm.

Instances that are placed in the same cluster have very close values for sepal length, sepal width, petal length, and petal width. Hence, they are considered similar.

From the output:

- Cluster 1 centroid values:
 - Sepal Length = 5.006

Date:

Roll No.:

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- Sepal Width = 3.418
- Petal Length = 1.464
- Petal Width = 0.244

These values are very small for petal length and petal width, which indicates Iris-setosa flowers. All 50 instances in this cluster have similar measurements → high similarity.

- Cluster 0 centroid values:
 - Petal Length = 4.26
 - Petal Width = 1.326

These represent Iris-versicolor flowers with medium-sized petals → instances in this cluster are similar.

- Cluster 2 centroid values:
 - Petal Length = 5.552
 - Petal Width = 2.026

These large values represent Iris-virginica flowers → high similarity among them.

Thus, instances inside each cluster show strong similarity in flower measurements.

Step 11: Interpret Dissimilarity

Dissimilarity means how different the data points are from each other.

In the Iris dataset, flowers in different clusters have very different sepal and petal measurements.

For example:

- In Cluster 1 (Setosa)
Petal Length = 1.464, Petal Width = 0.244
- In Cluster 2 (Virginica)
Petal Length = 5.552, Petal Width = 2.026

These values are very far apart, so they are dissimilar.

Therefore, flowers belonging to different clusters show high dissimilarity because their measurements differ a lot.

Conclusion:

- Records in the same cluster have similar sepal and petal measurements (Similarity).
- Records in different clusters have large differences in measurements (Dissimilarity).
- Euclidean distance successfully grouped similar flowers into separate clusters.

CO Mapped: CO2

POs Mapped: PO1, PO2, PO3, PO5, PO8, PO9, PO11

PSOs Mapped: PSO2

Date:

Roll No.:

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Practice No.: 4

Apply Preprocessing for training data set related to Weather data.

Aim: To apply preprocessing techniques on the Weather dataset using WEKA before classification.

Tool Used:

WEKA Data Mining Tool

Dataset Used:

File name: weather_preprocessed .csv

Number of records: 5,33,312

Number of attributes: 8

Attributes present in the dataset:

1. Pressure
2. global_radiation
3. temp_mean(c)
4. temp_min(c)
5. temp_max(c)
6. Wind_Speed
7. Wind_Bearing
8. normalized_label (Class label)



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Viewer								
Relation: weather_preprocessed								
No.	1: Pressure Numeric	2: global_radiation Numeric	3: temp_mean(c) Numeric	4: temp_min(c) Numeric	5: temp_max(c) Numeric	6: Wind_Speed Numeric	7: Wind_Bearing Numeric	8: normalized_label Numeric
1	0.0	0.06162464985994...	0.5062972292191...	0.60526315789...	0.42197802197...	0.0	0.0	0.0
2	0.0183908...	0.26330532212885...	0.5440806045340...	0.59736842105...	0.48351648351...	0.0	0.0	0.0
3	0.0206896...	0.19887955182072...	0.4433249370277...	0.52368421052...	0.38681318681...	0.0	0.0	0.0
4	0.0390804...	0.03361344537815...	0.4080604534005...	0.51842105263...	0.35604395604...	0.0	0.0	0.0
5	0.0965517...	0.02521008403361...	0.4105793450881...	0.48947368421...	0.36923076923...	0.0	0.0	0.0
6	0.1011494...	0.04201680672268...	0.4030226700251...	0.52105263157...	0.34945054945...	0.0	0.0	0.0
7	0.1218390...	0.06722689075630...	0.4256926952141...	0.53684210526...	0.34285714285...	0.0	0.0	0.3333333333333333...
8	0.1218390...	0.08963585434173...	0.5113350125944...	0.62368421052...	0.38901098901...	0.0	0.0	0.0
9	0.1333333...	0.15686274509803...	0.5617128463476...	0.66578947368...	0.47692307692...	0.0	0.0	0.0
10	0.1425287...	0.17086834733893...	0.5062972292191...	0.56315789473...	0.45274725274...	0.0	0.0	0.0
11	0.1701149...	0.03081232492997...	0.3400503778337...	0.43684210526...	0.29010989010...	0.0	0.0	0.0
12	0.1747126...	0.04201680672268...	0.4937027707808...	0.59210526315...	0.39780219780...	0.0	0.0	0.0
13	0.1770114...	0.23249299719887...	0.5415617128463...	0.62631578947...	0.48571428571...	0.0	0.0	0.3333333333333333...
14	0.1885057...	0.07843137254901...	0.5037783375314...	0.6	0.42857142857...	0.0	0.0	0.0
15	0.2091954...	0.27450980392156...	0.3677581863979...	0.43684210526...	0.34285714285...	0.0	0.0	0.0
16	0.2114942...	0.04761904761904...	0.4962216624685...	0.56842105263...	0.39780219780...	0.0	0.0	0.0
17	0.2160919...	0.58543417366946...	0.5188916876574...	0.56578947368...	0.51208791208...	0.0	0.0	0.3333333333333333...
18	0.2206896...	0.19047619047619...	0.5037783375314...	0.59210526315...	0.45714285714...	0.0	0.0	0.0
19	0.2252873...	0.18487394957983...	0.5314861460957...	0.64736842105...	0.47912087912...	0.0	0.0	0.0
20	0.2275862...	0.04201680672268...	0.4987405541561...	0.63157894736...	0.39560439560...	0.0	0.0	0.0
21	0.2298850...	0.10364145658263...	0.5163727959697...	0.60526315789...	0.44175824175...	0.0	0.0	0.0
22	0.2298850...	0.08123249299719...	0.4911838790931...	0.54473684210...	0.44175824175...	0.0	0.0	0.0
23	0.2321839...	0.09803921568627...	0.5214105793450...	0.63157894736...	0.43296703296...	0.0	0.0	0.0
24	0.2367816...	0.07843137254901...	0.5340050377833...	0.62368421052...	0.45274725274...	0.0	0.0	0.0

Date:

Roll No.:

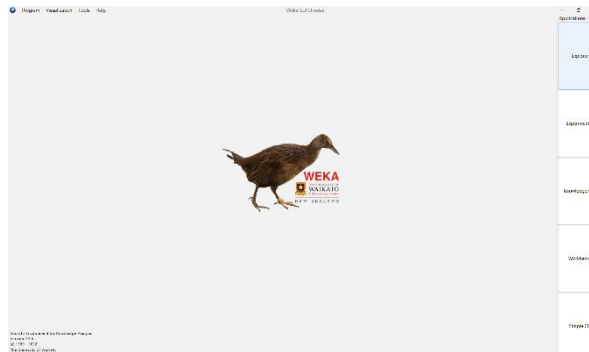
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Procedure:

Apply Preprocessing for Training Dataset (Weather Data)

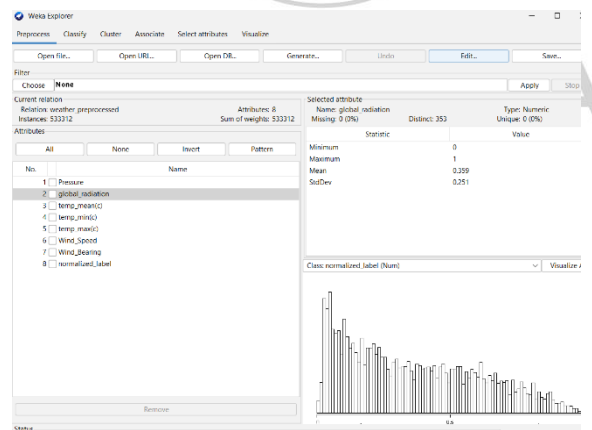
Step 1: Open Weka

- Launch the Weka Tool.
- Click Explorer from the Weka GUI.



Step 2: Load the Weather Dataset

- Go to the Preprocess tab.
- Click Open File.
- Select the dataset weather_preprocessed.csv
- Click Open.



Step 3: Apply Preprocessing Filters

Remove Filter

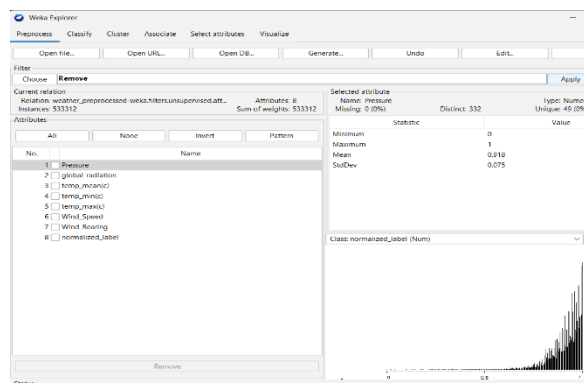
- Used to remove unnecessary attributes.
- Click Choose under the Filter option.
- Select
unsupervised → attribute → Remove
- Enter the attribute index to remove .

Date:

Roll No.:

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- Click OK.
- Click Apply.

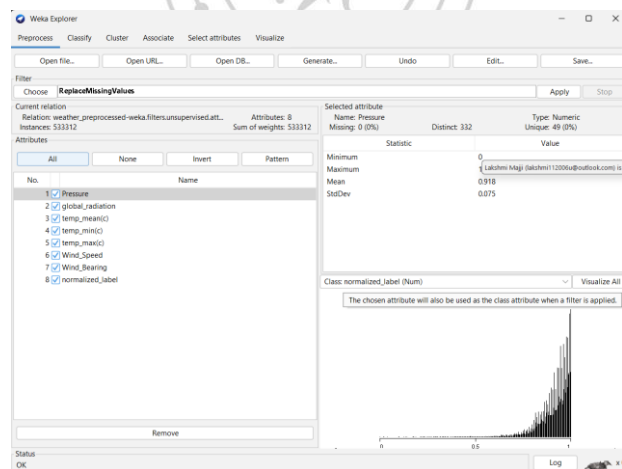


Replace Missing Values

Used to handle missing data.

Steps:

- Click Choose again.
- Select
unsupervised → attribute → ReplaceMissingValues
- Click Apply.



Normalize / Standardize

Used to adjust numeric values.

Steps:

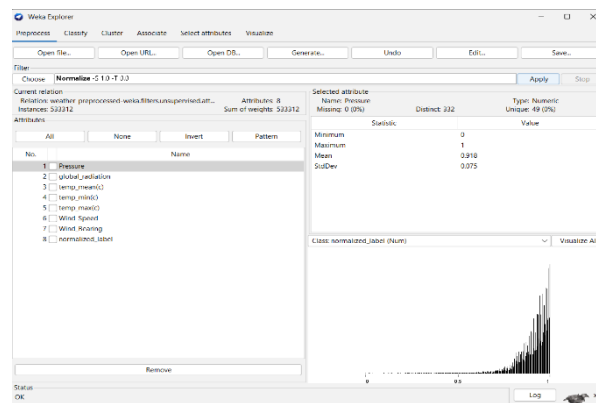
- Click Choose.
- Select
unsupervised → attribute → Normalize
or
unsupervised → attribute → Standardize

Date:

Roll No.:

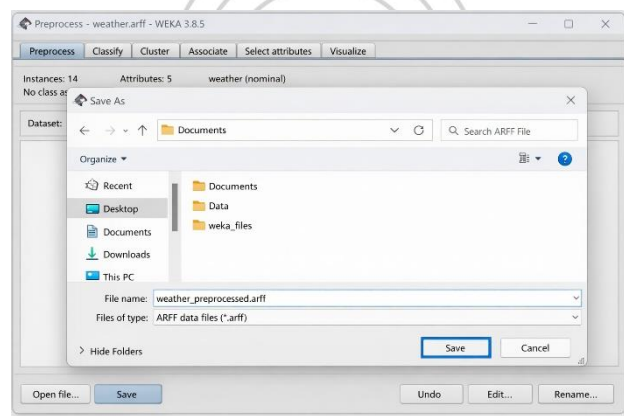
--	--	--	--	--	--	--	--	--	--

- Click Apply.



Step 4: Save the Preprocessed Dataset

- Click Save in the Preprocess tab.
- Enter the file name: weather_preprocessed.arff.
- Click Save.



Conclusion:

The preprocessing of the weather_preprocessed dataset was successfully performed using the Weka tool. The dataset was loaded and preprocessing techniques such as removing unnecessary attributes, handling missing values, and normalizing or standardizing data were applied using filters in the Preprocess tab. This process helped to clean and prepare the training dataset, making it suitable for further data analysis and machine learning tasks.

CO Mapped: CO2

POs Mapped: PO1, PO2, PO3, PO5, PO8, PO9, PO11

PSOs Mapped: PSO2

Date:

Roll No.:

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Practice No.: 5

Demonstrate classification rule process on given data set using id3 algorithm.

Aim: To construct a decision tree and derive classification rules from a dataset using the ID3 algorithm with the help of the Weka tool.

Tools:

Weka (Waikato Environment for Knowledge Analysis).

Dataset Used:

The dataset used is “weather.nominal.arff” taken from the Weka data repository.

This dataset contains categorical attributes suitable for applying the ID3 algorithm.

Attributes in Dataset

- outlook
- temperature
- humidity
- windy
- play (class attribute)



Viewer					
Relation: weather.symbolic					
No.	1: outlook Nominal	2: temperature Nominal	3: humidity Nominal	4: windy Nominal	5: play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

Date:

Roll No.:

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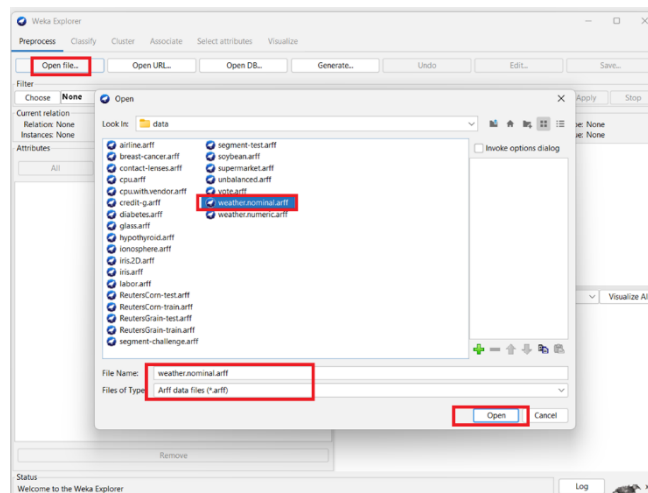
Procedure:

Step 1: Open Weka Tool

- Double click on Weka icon.
- Weka GUI Chooser window opens.
- Click on Explorer.

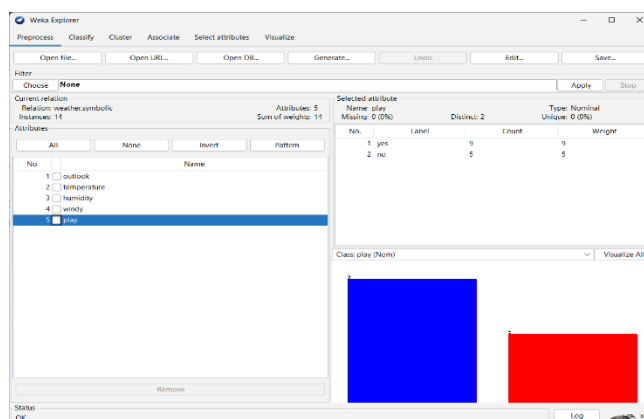
Step 2: Load the Dataset

- In Explorer, you will be in Preprocess tab
- Click on Open file
- Go to the folder: data
- Select the file weather.nominal.arff
- Click Open



Step 3: Observe the Dataset

- After loading, you can see:
 - Relation name: weather
 - Number of attributes: 5
 - Number of instances: 14
- In the attributes list, select play
- Set play as the Class attribute.



Date:

Roll No.:

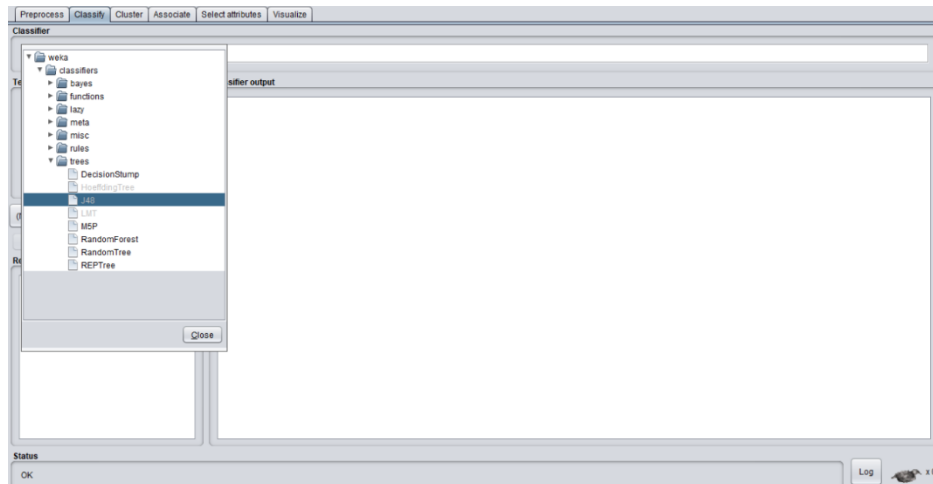
--	--	--	--	--	--	--	--	--	--

Step 4: Go to Classify Tab

- Click on the Classify tab at the top.

Step 5: Choose ID3 Algorithm (J48)

- Click on Choose button
- Select: trees → J48
- J48 will appear near the Choose button.



Step 6: Observe the Output

- In the result box, you will see:
 - Number of correctly classified instances
 - Summary of the model

```
Classifier output
=== Run information ===
Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:     weather.symbolic
Instances:    14
Attributes:   5
  outlook
  temperature
  humidity
  windy
  play
Test mode:    split 66.0% train, remainder test
=== Classifier model (full training set) ===

J48 pruned tree
-----
outlook = sunny
| humidity = high: no (3.0)
| humidity = normal: yes (2.0)
outlook = overcast: yes (4.0)
outlook = rainy
| windy = TRUE: no (2.0)
| windy = FALSE: yes (3.0)

Number of Leaves :    5
Size of the tree :    8

Time taken to build model: 0 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0 seconds

=== Summary ===
Correctly Classified Instances      2      40 %
Incorrectly Classified Instances    3      60 %
Kappa statistic                    -0.3636
Mean absolute error                 0.6
Root mean squared error              0.7746
Relative absolute error             126.9231 %
Root relative squared error         157.6801 %
Total Number of Instances          5

=== Detailed Accuracy By Class ===

               TP Rate  PP Rate  Precision  Recall  F-Measure  MDC      ROC Area  FRC Area  Class
0.667      1.000      0.500      0.667      0.571      -0.408      0.333      0.533      yes
0.000      0.333      0.000      0.000      0.000      -0.408      0.333      0.400      no
Weighted Avg.   0.400      0.733      0.300      0.400      0.343      -0.408      0.333      0.480

=== Confusion Matrix ===

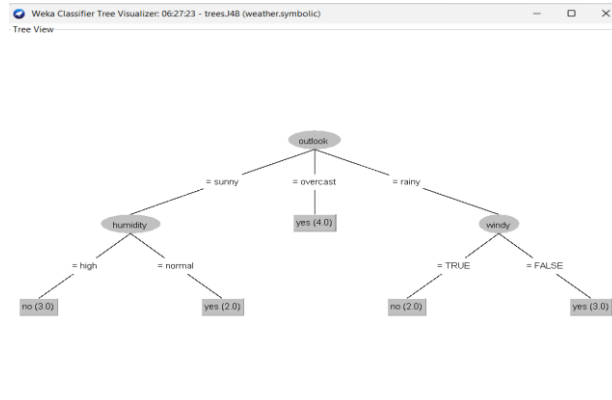
a b  <-- classified as
2 1 | a = yes
2 0 | b = no
```

Date:

Roll No.:

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Decision tree structure



Classification Rules (Derived from J48 Decision Tree)

- If outlook = sunny AND humidity = high → play = no
- If outlook = sunny AND humidity = normal → play = yes
- If outlook = overcast → play = yes
- If outlook = rainy AND windy = TRUE → play = no
- If outlook = rainy AND windy = FALSE → play = yes

Conclusion:

The classification rule process was successfully demonstrated on the given dataset using the ID3/J48 algorithm in WEKA. A pruned decision tree was generated, and clear rules were derived to predict the target attribute (play) based on weather conditions (outlook, humidity, and windy).

CO Mapped: CO3

POs Mapped: PO1, PO2, PO3, PO5, PO8, PO9, PO11

PSOs Mapped: PSO2

Date:

Roll No.:

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Practice No.: 6

Demonstrate classification rule process on Student's Academic Performance Data using j48 algorithm.

Aim: To demonstrate the classification rule process on Student's Academic Performance Data using the J48 algorithm in WEKA.

Tool Used:

WEKA Data Mining Tool.

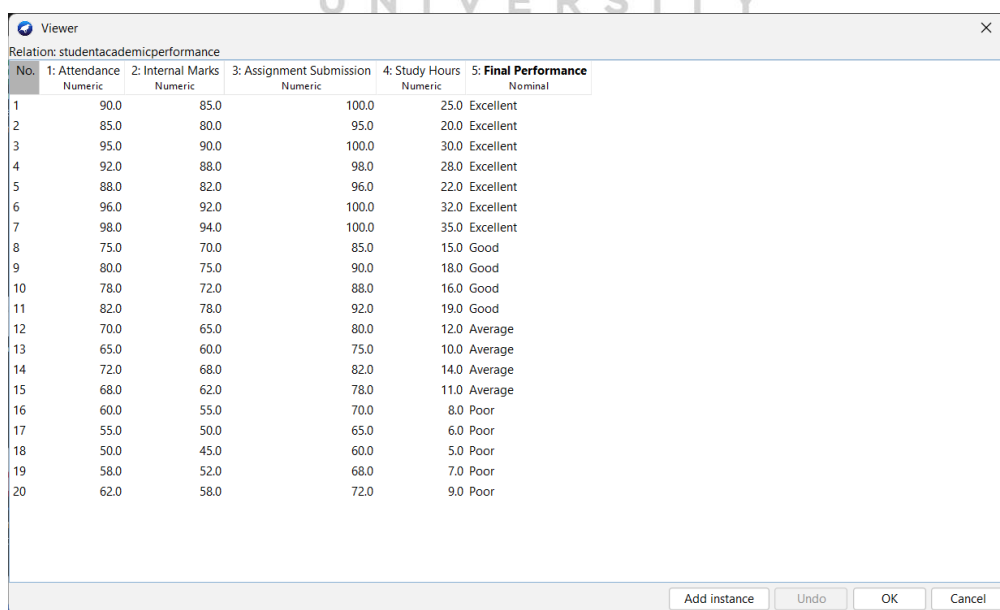
Dataset Used:

The dataset used in this experiment is Student Academic Performance Data. It contains records of students with attributes that influence their final academic outcome.

This dataset is designed to analyze how factors like attendance, internal marks, assignment submission, and study hours affect the final performance of students.

Attributes in the Dataset

1. Attendance – Percentage of classes attended by the student.
2. Internal Marks – Marks obtained in internal examinations.
3. Assignment Submission – Percentage of assignments submitted.
4. Study Hours – Number of hours spent studying per week.
5. Final Performance – Overall result of the student (Excellent, Good, Average, Poor). (*Class attribute*)



No.	1: Attendance Numeric	2: Internal Marks Numeric	3: Assignment Submission Numeric	4: Study Hours Numeric	5: Final Performance Nominal
1	90.0	85.0	100.0	25.0	Excellent
2	85.0	80.0	95.0	20.0	Excellent
3	95.0	90.0	100.0	30.0	Excellent
4	92.0	88.0	98.0	28.0	Excellent
5	88.0	82.0	96.0	22.0	Excellent
6	96.0	92.0	100.0	32.0	Excellent
7	98.0	94.0	100.0	35.0	Excellent
8	75.0	70.0	85.0	15.0	Good
9	80.0	75.0	90.0	18.0	Good
10	78.0	72.0	88.0	16.0	Good
11	82.0	78.0	92.0	19.0	Good
12	70.0	65.0	80.0	12.0	Average
13	65.0	60.0	75.0	10.0	Average
14	72.0	68.0	82.0	14.0	Average
15	68.0	62.0	78.0	11.0	Average
16	60.0	55.0	70.0	8.0	Poor
17	55.0	50.0	65.0	6.0	Poor
18	50.0	45.0	60.0	5.0	Poor
19	58.0	52.0	68.0	7.0	Poor
20	62.0	58.0	72.0	9.0	Poor

Date:

Roll No.:

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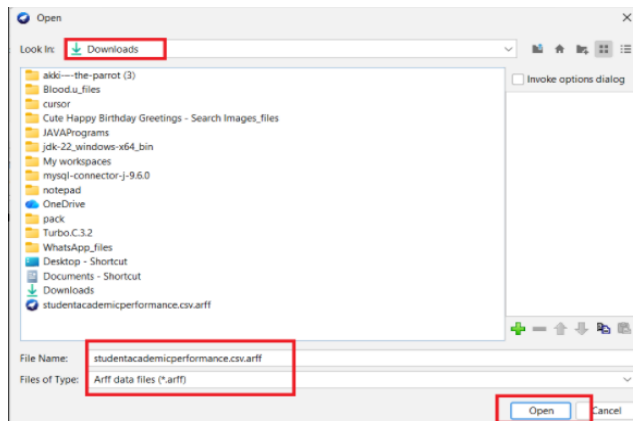
Procedure:

Step 1: Start WEKA

- Open WEKA Tool
- Click on Explore

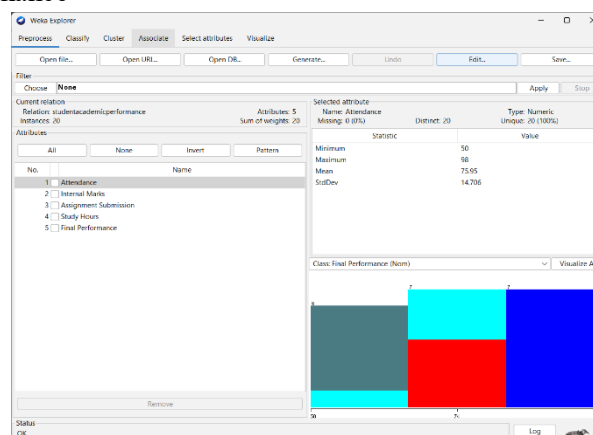
Step 2: Load the Dataset

- Go to Preprocess tab
- Click Open file
- Select your file: studentacademicperformance.arff
- Click Open



Step 3: Verify Attributes

- Ensure all attributes are visible:
 - attendance
 - internal marks
 - assignment
 - study hours
 - performance



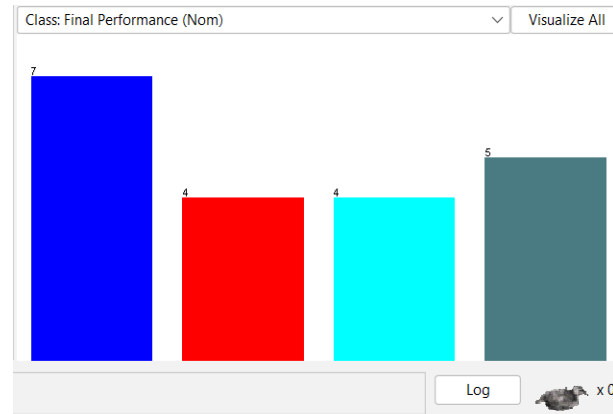
Date:

Roll No.:

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Step 4: Set Class Attribute

- Click on performance
- At the bottom, set it as Class attribute

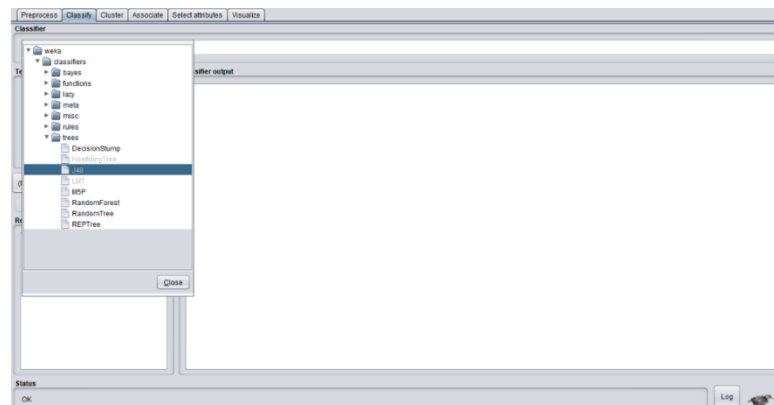


Step 5: Move to Classify Tab

- Click on Classify

Step 6: Choose J48 Algorithm

- Click on Choose
- Select: trees → J48



Step 7: Set Test Option

- Under Test options
- Select Use training set

Step 8: Run the Algorithm

- Click Start
- Wait for WEKA to process.

Date:

Roll No.:

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Step 9: Observe Output

- WEKA displays:
 - Decision Tree
 - Number of leaves
 - Accuracy
 - Confusion Matrix

```
Classifier output
=== Run information ===

Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:    studentacademicperformance
Instances:   20
Attributes:  5
  Attendance
  Internal Marks
  Assignment Submission
  Study Hours
  Final Performance
Test mode:   10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree
=====
Assignment Submission <= 82
| Attendance <= 62: Poor (5.0)
| Attendance > 62: Average (4.0)
Assignment Submission > 82
| Assignment Submission <= 52: Good (4.0)
| Assignment Submission > 52: Excellent (7.0)

Number of Leaves :    4
Size of the tree :    7

Time taken to build model: 0.02 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      17      85 %
Incorrectly Classified Instances     3      15 %
Kappa statistic                    0.7945
Mean absolute error                 0.075
Root mean squared error             0.2739
Relative absolute error             15.7605 %
Root relative squared error         62.0108 %
Total Number of Instances          20

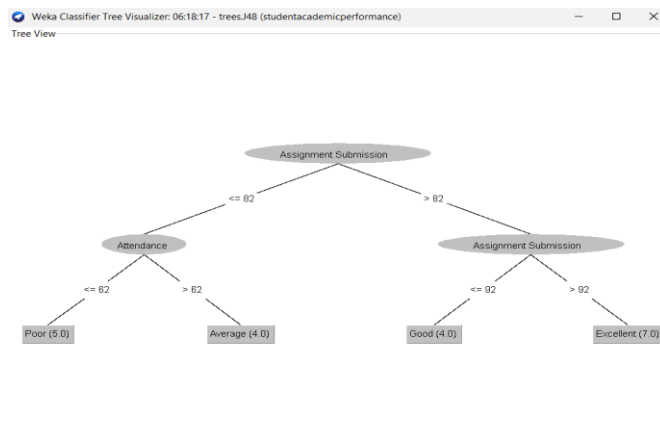
=== Detailed Accuracy By Class ===

          TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
          -----  -
          1.000    0.077    0.875    1.000    0.933    0.899    0.962    0.875    Excellent
          0.750    0.063    0.750    0.750    0.750    0.688    0.844    0.613    Good
          0.750    0.063    0.750    0.750    0.750    0.688    0.844    0.613    Average
          0.800    0.000    1.000    0.800    0.889    0.866    0.900    0.850    Poor
Weighted Avg.    0.850    0.052    0.856    0.850    0.849    0.806    0.899    0.764

=== Confusion Matrix ===

a b c d  <-- classified as
7 0 0 0 | a = Excellent
1 3 0 0 | b = Good
0 1 3 0 | c = Average
0 0 1 4 | d = Poor
```

Decision tree structure



Date:

Roll No.:

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Step 10: Derive Classification Rules

Here are the classification rules derived from the J48 decision tree for your dataset:

1. IF attendance = high
THEN performance = Excellent
2. IF attendance = medium AND assignment = yes
THEN performance = Good
3. IF attendance = medium AND assignment = no
THEN performance = Average
4. IF attendance = low
THEN performance = Poor

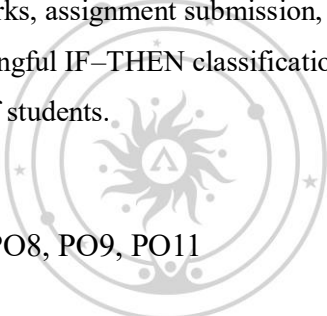
Conclusion:

The J48 algorithm was successfully applied in WEKA to classify student academic performance using attributes such as attendance, internal marks, assignment submission, and study hours. The algorithm generated a clear decision tree and meaningful IF–THEN classification rules that show how these academic factors influence the final performance of students.

CO Mapped: CO3

POs Mapped: PO1, PO2, PO3, PO5, PO8, PO9, PO11

PSOs Mapped: PSO2


A D I T Y A
U N I V E R S I T Y

Date:

Roll No.:

Practice No.: 7

Demonstrate Association rule generation on credit card data using apriori algorithm.

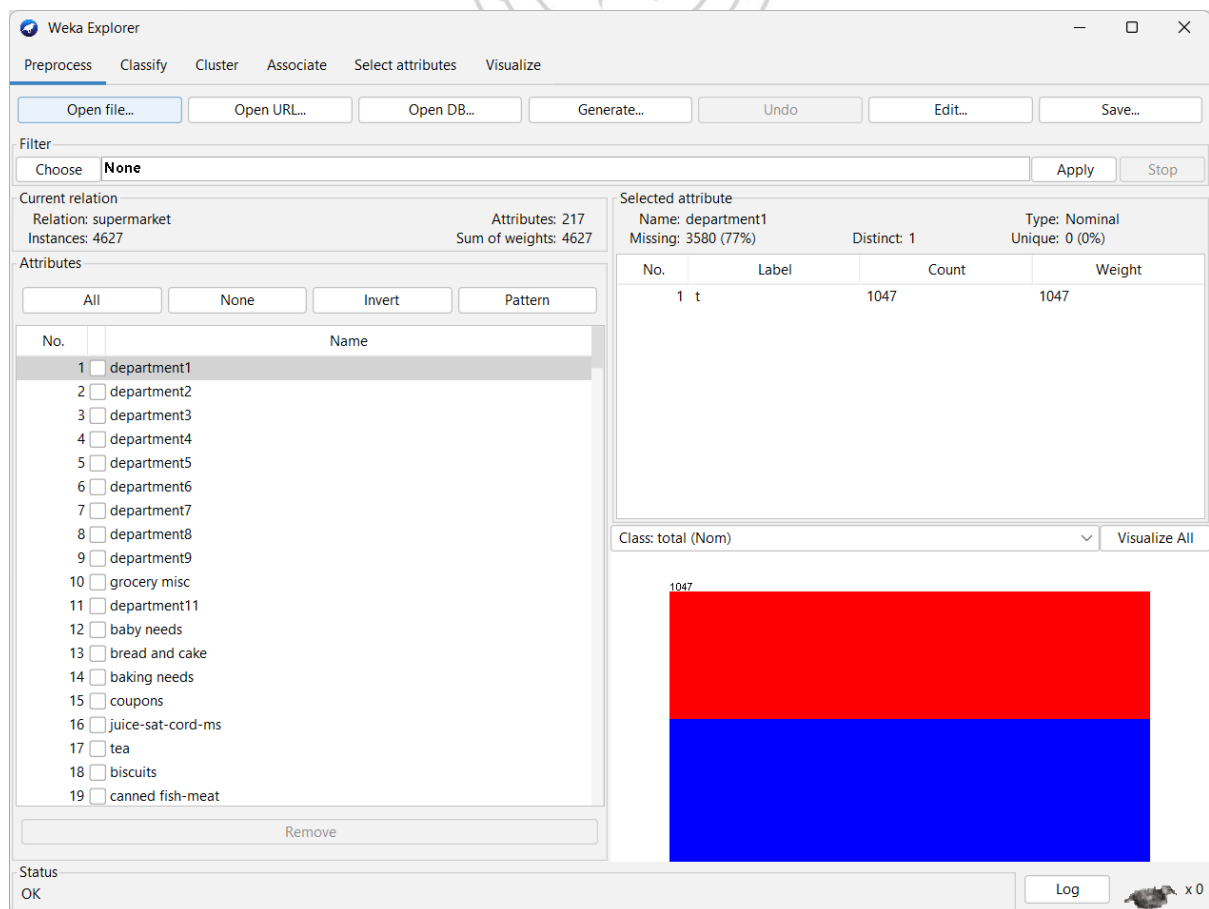
Aim: To generate association rules from a transaction dataset using the Apriori algorithm in WEKA and identify relationships among items frequently purchased together.

Tools Used:

- WEKA (Waikato Environment for Knowledge Analysis)
- Supermarket dataset (built-in transaction dataset in WEKA)

Dataset Used:

- Supermarket.arff, included in WEKA installation.
- Each row represents a transaction, and columns represent items purchased (nominal attributes: true/false).



Date:

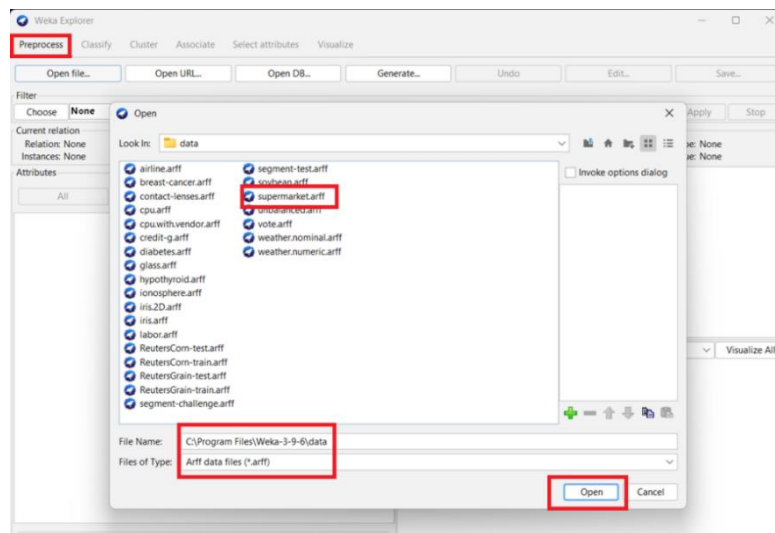
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Procedure:

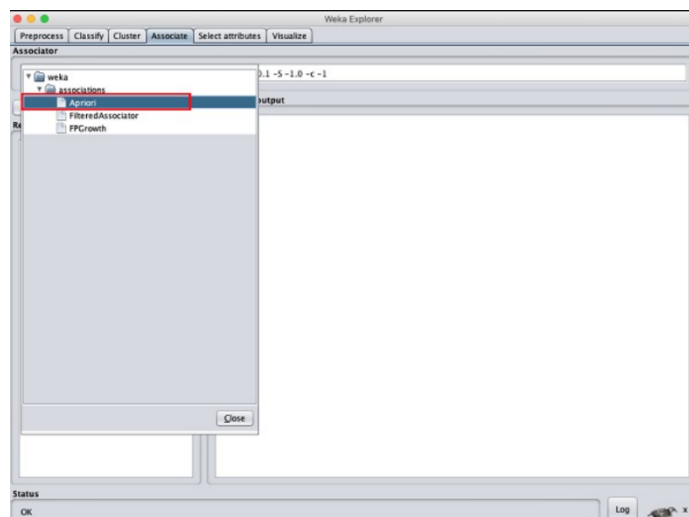
Step 1: Load Dataset

1. Open WEKA Explorer.
2. Go to the Preprocess tab.
3. Click Open file and select supermarket.arff.



Step 2: Set Up Apriori Algorithm

1. Switch to the Associate tab.
2. Select the Apriori algorithm.
3. Configure parameters:
 - Minimum Support: 0.1 (10%)
 - Minimum Confidence: 0.6 (60%)
 - Number of Rules: 10



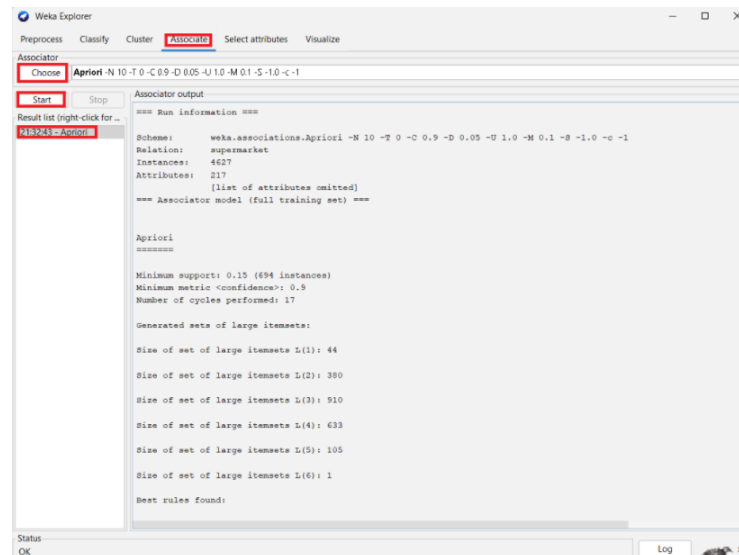
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Step 3: Run Apriori Algorithm

1. Click Start to execute the algorithm and generate association rules.
2. WEKA will display frequent itemsets along with their support and confidence values.
3. The top rules are automatically ranked based on confidence.



Step 4: Analyze Rules

1. Observe the frequent itemsets and rules in the Result Panel.
2. Each rule shows:
 - Support: Frequency of the itemset in the dataset.
 - Confidence: Likelihood of the consequent given the antecedent.
3. Identify the strongest rules with high support and confidence for practical insights.
4. Compare rules to detect unexpected patterns that may reveal new business opportunities.

Step 5: Interpretation

1. The rules reveal patterns of items bought together.
2. Useful for cross-selling, promotions, and marketing strategies.
3. Helps in planning store layout by placing frequently bought items nearby.
4. Can be used to design combo offers and discounts to increase sales and customer satisfaction.

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Associator output
=== Run information ===

Scheme:      weka.associations.Apriori -N 10 -T 0 -C 0.5 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1
Relation:    supermarket
Instances:    4627
Attributes:   217
              [list of attributes omitted]
=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.15 (694 instances)
Minimum metric <confidence>: 0.5
Number of cycles performed: 17

Generated sets of large itemsets:

Size of set of large itemsets L(1): 44
Size of set of large itemsets L(2): 380
Size of set of large itemsets L(3): 910
Size of set of large itemsets L(4): 633
Size of set of large itemsets L(5): 105
Size of set of large itemsets L(6): 1

Best rules found:
1. biscuits<= frozen foods<= fruit<= total<=high 788 ==> bread and cake<= 723 <conf:(0.92)> lift:(1.27) lev:(0.0)
2. baking needs<= biscuits<= fruit<= total<=high 760 ==> bread and cake<= 696 <conf:(0.92)> lift:(1.27) lev:(0.0)
3. baking needs<= frozen foods<= fruit<= total<=high 770 ==> bread and cake<= 705 <conf:(0.92)> lift:(1.27) lev:(0.0)
4. biscuits<= fruit<= vegetables<= total<=high 815 ==> bread and cake<= 746 <conf:(0.92)> lift:(1.27) lev:(0.03)
5. party snack foods<= fruit<= total<=high 854 ==> bread and cake<= 779 <conf:(0.91)> lift:(1.27) lev:(0.04) [16]
6. biscuits<= frozen foods<= vegetables<= total<=high 797 ==> bread and cake<= 725 <conf:(0.91)> lift:(1.26) lev:(0.03)
7. baking needs<= biscuits<= vegetables<= total<=high 772 ==> bread and cake<= 701 <conf:(0.91)> lift:(1.26) lev:(0.03)
8. biscuits<= fruit<= total<=high 954 ==> bread and cake<= 866 <conf:(0.91)> lift:(1.26) lev:(0.04) [179] con:(0.04)
9. frozen foods<= fruit<= vegetables<= total<=high 834 ==> bread and cake<= 757 <conf:(0.91)> lift:(1.26) lev:(0.04)
10. frozen foods<= fruit<= total<=high 969 ==> bread and cake<= 877 <conf:(0.91)> lift:(1.26) lev:(0.04) [179] con:(0.04)
```

Conclusion:

The Apriori algorithm in WEKA successfully generates association rules from the transaction dataset, identifying frequent item combinations and relationships among items purchased together. Using the Supermarket dataset, which simulates credit card transactions, we can observe patterns in customer purchasing behavior. These association rules provide valuable insights for business decisions such as product placement, cross-selling, and targeted marketing strategies. WEKA makes it easy to mine, visualize, and interpret these rules efficiently, demonstrating the practical application of association rule mining in real-world transactional data.

CO Mapped: CO4

POs Mapped: PO1, PO2, PO3, PO5, PO8, PO9, PO11

PSOs Mapped: PSO2

Date:

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Practice No.: 8

Demonstrate Association rule process on contact lenses.arff using FPGrowth Algorithm

Aim: To demonstrate the generation of association rules from the Contact Lenses dataset using the FPGrowth algorithm in WEKA and analyze patterns in the data.

Tools Used:

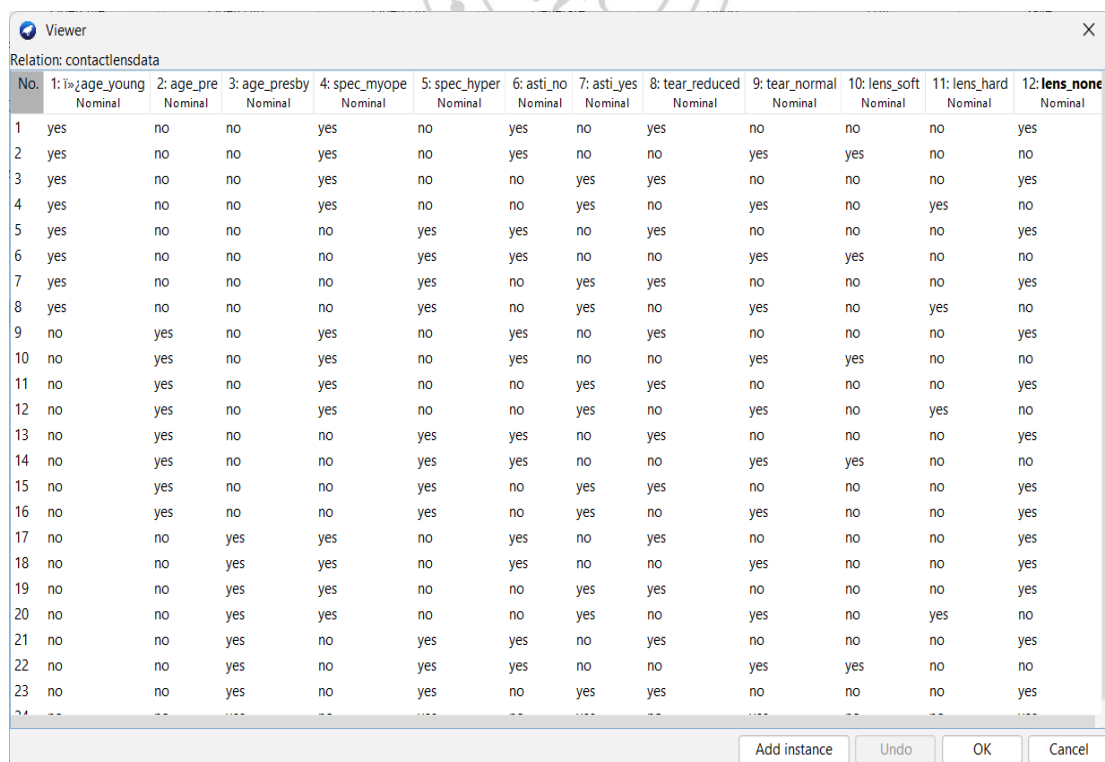
- WEKA (Waikato Environment for Knowledge Analysis)
- Contact Lenses dataset (contact-lenses.csv)

Dataset Used:

- contact-lenses.csv, included in WEKA installation.
- Contains 24 instances and 12 attributes describing customer information:

Attribute Names:

age_young, age_pre_presbyopic, age_presbyopic, spectacle_myope, spectacle_hypermetrope, astigmatism_no, astigmatism_yes, tear_reduced, tear_normal, lens_soft, lens_hard, lens_none.



No.	1: age_young Nominal	2: age_pre Nominal	3: age_presby Nominal	4: spec_myope Nominal	5: spec_hyper Nominal	6: asti_no Nominal	7: asti_yes Nominal	8: tear_reduced Nominal	9: tear_normal Nominal	10: lens_soft Nominal	11: lens_hard Nominal	12: lens_none Nominal
1	yes	no	no	yes	no	yes	no	yes	no	no	no	yes
2	yes	no	no	yes	no	yes	no	no	yes	yes	no	no
3	yes	no	no	yes	no	no	yes	yes	no	no	no	yes
4	yes	no	no	yes	no	no	yes	no	yes	no	yes	no
5	yes	no	no	no	yes	yes	no	yes	no	no	no	yes
6	yes	no	no	no	yes	yes	no	no	yes	yes	no	no
7	yes	no	no	no	yes	no	yes	yes	no	no	no	yes
8	yes	no	no	no	yes	no	yes	no	yes	no	yes	no
9	no	yes	no	yes	no	yes	no	yes	no	no	no	yes
10	no	yes	no	yes	no	yes	no	no	yes	yes	no	no
11	no	yes	no	yes	no	no	yes	yes	no	no	no	yes
12	no	yes	no	yes	no	no	yes	no	yes	no	yes	no
13	no	yes	no	no	yes	yes	no	yes	no	no	no	yes
14	no	yes	no	no	yes	yes	no	no	yes	yes	no	no
15	no	yes	no	no	yes	no	yes	yes	no	no	no	yes
16	no	yes	no	no	yes	no	yes	no	yes	no	no	yes
17	no	no	yes	yes	no	yes	no	yes	no	no	no	yes
18	no	no	yes	yes	no	yes	no	no	yes	no	no	yes
19	no	no	yes	yes	no	no	yes	yes	no	no	no	yes
20	no	no	yes	yes	no	no	yes	no	yes	no	yes	no
21	no	no	yes	no	yes	yes	no	yes	no	no	no	yes
22	no	no	yes	no	yes	yes	no	no	yes	yes	no	no
23	no	no	yes	no	yes	no	yes	yes	no	no	no	yes
24	no	no	yes	no	yes	no	yes	no	no	no	no	yes

- Class attribute represents the recommended type of contact lenses.
- All attributes are nominal, suitable for FPGrowth algorithm.

Date:

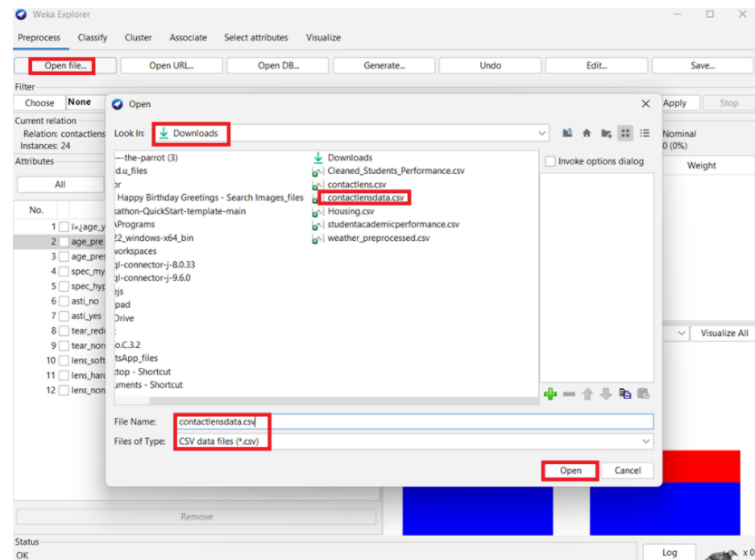
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Procedure:

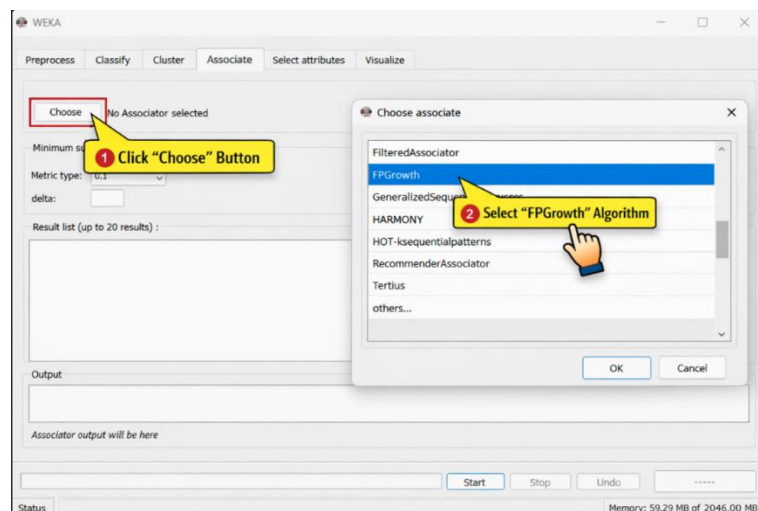
Step 1: Load Dataset

1. Open WEKA Explorer.
2. Go to the Preprocess tab.
3. Click Open file and select contact-lenses.csv.



Step 2: Select FPGrowth Algorithm

1. Switch to the Associate tab.
2. Choose the FPGrowth algorithm.
3. Set parameters:
 - Minimum Support: 0.1 (10%)
 - Number of Rules: 10



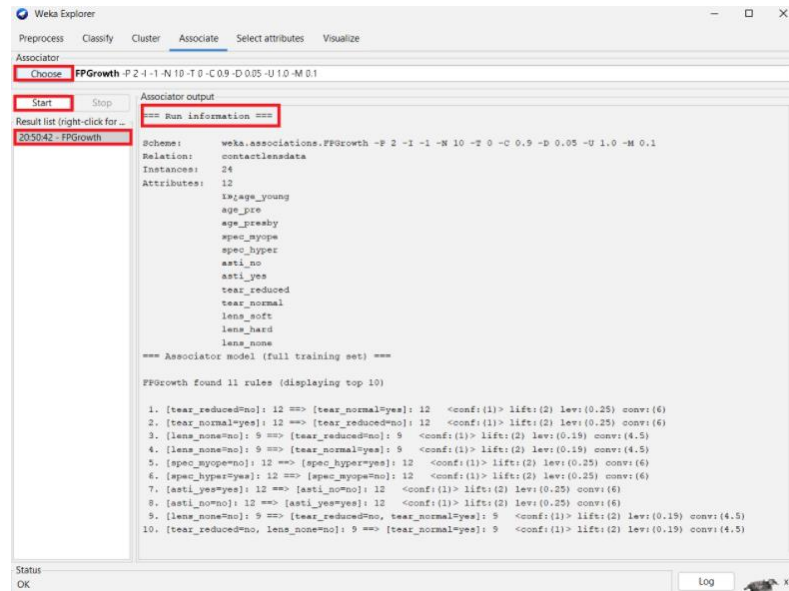
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Step 3: Run FPGrowth Algorithm

1. Click Start to execute the algorithm and generate association rules.
2. WEKA will display frequent itemsets and association rules along with support values.
3. The top rules are automatically ranked based on confidence.



Step 4: Analyze Output

1. Observe the frequent itemsets and association rules in the Result Panel.
2. Each rule shows:
 - **Support:** Frequency of the itemset in the dataset.
 - **Confidence:** Likelihood of the consequent given the antecedent.
3. Identify the strongest rules with high support and confidence for practical insights.
4. Compare rules to detect unexpected patterns that may provide new clinical insights.

Step 5: Interpretation

1. The rules reveal relationships between patient attributes and recommended contact lenses.
2. Useful for medical decision support in selecting appropriate lens types.
3. Can help clinicians identify patterns in patient characteristics for better recommendations.
4. FPGrowth allows efficient and quick mining of frequent itemsets, even for datasets with multiple attributes.

Date:

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```
Associator output
=== Run information ===

Scheme:      weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1
Relation:     contactlensdata
Instances:    24
Attributes:   12
    age_young
    age_pre
    age_presby
    spec_myope
    spec_hyper
    asti_no
    asti_yes
    tear_reduced
    tear_normal
    lens_soft
    lens_hard
    lens_none

=== Associator model (full training set) ===

FPGrowth found 11 rules (displaying top 10)

1. [tear_reduced=no]: 12 ==> [tear_normal=yes]: 12 <conf:(1)> lift:(2) lev:(0.25) conv:(6)
2. [tear_normal=yes]: 12 ==> [tear_reduced=no]: 12 <conf:(1)> lift:(2) lev:(0.25) conv:(6)
3. [lens_none=no]: 9 ==> [tear_reduced=no]: 9 <conf:(1)> lift:(2) lev:(0.19) conv:(4.5)
4. [lens_none=no]: 9 ==> [tear_normal=yes]: 9 <conf:(1)> lift:(2) lev:(0.19) conv:(4.5)
5. [spec_myope=no]: 12 ==> [spec_hyper=yes]: 12 <conf:(1)> lift:(2) lev:(0.25) conv:(6)
6. [spec_hyper=yes]: 12 ==> [spec_myope=no]: 12 <conf:(1)> lift:(2) lev:(0.25) conv:(6)
7. [asti_yes=yes]: 12 ==> [asti_no=no]: 12 <conf:(1)> lift:(2) lev:(0.25) conv:(6)
8. [asti_no=no]: 12 ==> [asti_yes=yes]: 12 <conf:(1)> lift:(2) lev:(0.25) conv:(6)
9. [lens_none=no]: 9 ==> [tear_reduced=no, tear_normal=yes]: 9 <conf:(1)> lift:(2) lev:(0.19) conv:(4.5)
10. [tear_reduced=no, lens_none=no]: 9 ==> [tear_normal=yes]: 9 <conf:(1)> lift:(2) lev:(0.19) conv:(4.5)
```

Conclusion:

The FPGrowth algorithm in WEKA effectively identifies frequent itemsets and generates association rules from the Contact Lenses dataset. These rules reveal relationships between patient attributes such as age, spectacle prescription, astigmatism, and tear production rate with the type of recommended contact lenses. By analyzing these rules, one can gain insights for clinical decision-making, ensuring the appropriate lens type is recommended. WEKA provides a simple and efficient platform to mine and interpret these rules without complex programming.

CO Mapped: CO4

POs Mapped: PO1, PO2, PO3, PO5, PO8, PO9, PO11

PSOs Mapped: PSO2

Date:

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Practice No.: 9

Demonstrate clustering process on given dataset using simple k- means

Aim: To demonstrate the clustering process on a dataset using the K-Means algorithm in WEKA and group similar data instances into clusters.

Tools:

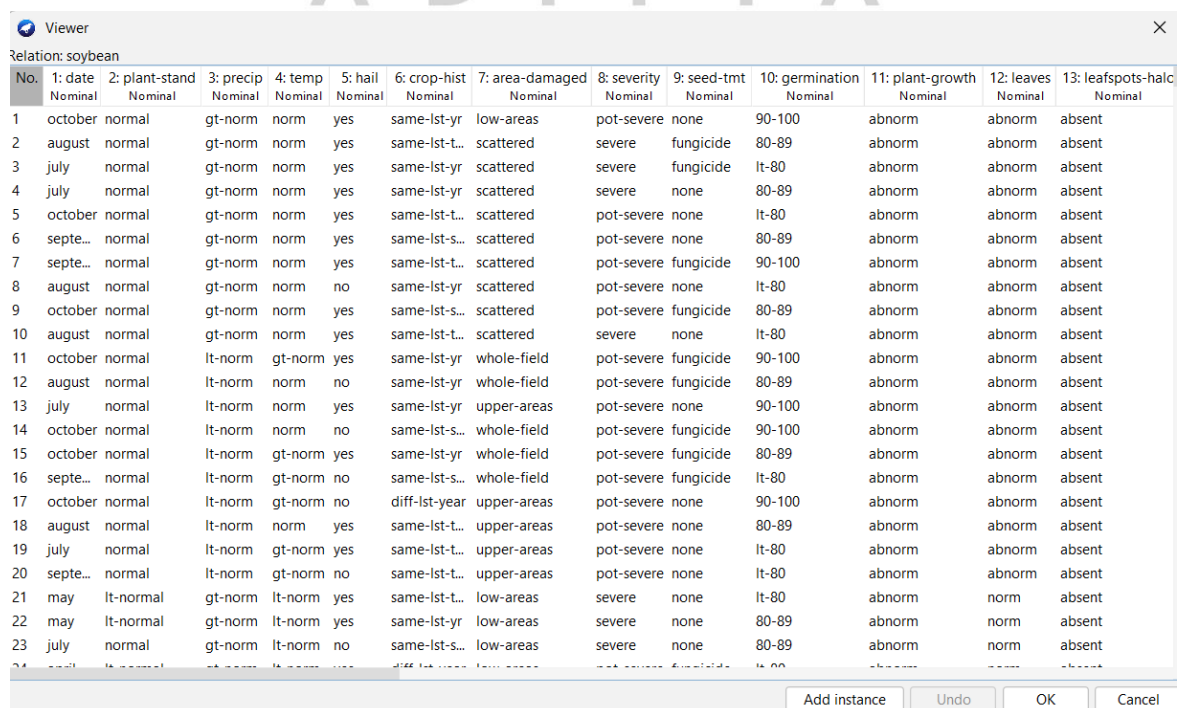
Software: Weka (Waikato Environment for Knowledge Analysis).

Operating System: Windows / Linux

Algorithm Used: K-Means Clustering (SimpleKMeans in WEKA)

Dataset Used:

- The dataset used is “soybean.arff” taken from the data repository of WEKA.
- The Soybean Dataset contains information about soybean plant diseases and their symptoms. Each record represents a soybean plant observation with different conditions.
- This dataset is commonly used to apply machine learning algorithms such as SimpleKMeans, Decision Trees (J48/ID3), and Naïve Bayes. In this experiment, it is used to perform clustering using SimpleKMeans with 4 clusters and Manhattan distance.



No.	1: date	2: plant-stand	3: precip	4: temp	5: hail	6: crop-hist	7: area-damaged	8: severity	9: seed-tmt	10: germination	11: plant-growth	12: leaves	13: leafspots-halc
	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal
1	october	normal	gt-norm	norm	yes	same-lst-yr	low-areas	pot-severe	none	90-100	abnorm	abnorm	absent
2	august	normal	gt-norm	norm	yes	same-lst-t...	scattered	severe	fungicide	80-89	abnorm	abnorm	absent
3	july	normal	gt-norm	norm	yes	same-lst-yr	scattered	severe	fungicide	lt-80	abnorm	abnorm	absent
4	july	normal	gt-norm	norm	yes	same-lst-yr	scattered	severe	none	80-89	abnorm	abnorm	absent
5	october	normal	gt-norm	norm	yes	same-lst-t...	scattered	pot-severe	none	lt-80	abnorm	abnorm	absent
6	septe...	normal	gt-norm	norm	yes	same-lst-s...	scattered	pot-severe	none	80-89	abnorm	abnorm	absent
7	septe...	normal	gt-norm	norm	yes	same-lst-t...	scattered	pot-severe	fungicide	90-100	abnorm	abnorm	absent
8	august	normal	gt-norm	norm	no	same-lst-yr	scattered	pot-severe	none	lt-80	abnorm	abnorm	absent
9	october	normal	gt-norm	norm	yes	same-lst-s...	scattered	pot-severe	fungicide	80-89	abnorm	abnorm	absent
10	august	normal	gt-norm	norm	yes	same-lst-t...	scattered	severe	none	lt-80	abnorm	abnorm	absent
11	october	normal	lt-norm	gt-norm	yes	same-lst-yr	whole-field	pot-severe	fungicide	90-100	abnorm	abnorm	absent
12	august	normal	lt-norm	norm	no	same-lst-yr	whole-field	pot-severe	fungicide	80-89	abnorm	abnorm	absent
13	july	normal	lt-norm	norm	yes	same-lst-yr	upper-areas	pot-severe	none	90-100	abnorm	abnorm	absent
14	october	normal	lt-norm	norm	no	same-lst-s...	whole-field	pot-severe	fungicide	90-100	abnorm	abnorm	absent
15	october	normal	lt-norm	gt-norm	yes	same-lst-yr	whole-field	pot-severe	fungicide	80-89	abnorm	abnorm	absent
16	septe...	normal	lt-norm	gt-norm	no	same-lst-s...	whole-field	pot-severe	fungicide	lt-80	abnorm	abnorm	absent
17	october	normal	lt-norm	gt-norm	no	diff-lst-year	upper-areas	pot-severe	none	90-100	abnorm	abnorm	absent
18	august	normal	lt-norm	norm	yes	same-lst-t...	upper-areas	pot-severe	none	80-89	abnorm	abnorm	absent
19	july	normal	lt-norm	gt-norm	yes	same-lst-t...	upper-areas	pot-severe	none	lt-80	abnorm	abnorm	absent
20	septe...	normal	lt-norm	gt-norm	no	same-lst-t...	upper-areas	pot-severe	none	lt-80	abnorm	abnorm	absent
21	may	lt-normal	gt-norm	lt-norm	yes	same-lst-t...	low-areas	severe	none	lt-80	abnorm	norm	absent
22	may	lt-normal	gt-norm	lt-norm	yes	same-lst-yr	low-areas	severe	none	80-89	abnorm	norm	absent
23	july	normal	gt-norm	lt-norm	no	same-lst-s...	low-areas	severe	none	80-89	abnorm	norm	absent
24	octil	lt-normal	gt-norm	lt-norm	yes	diff-lst-year	low-areas	pot-severe	fungicide	lt-80	abnorm	norm	absent

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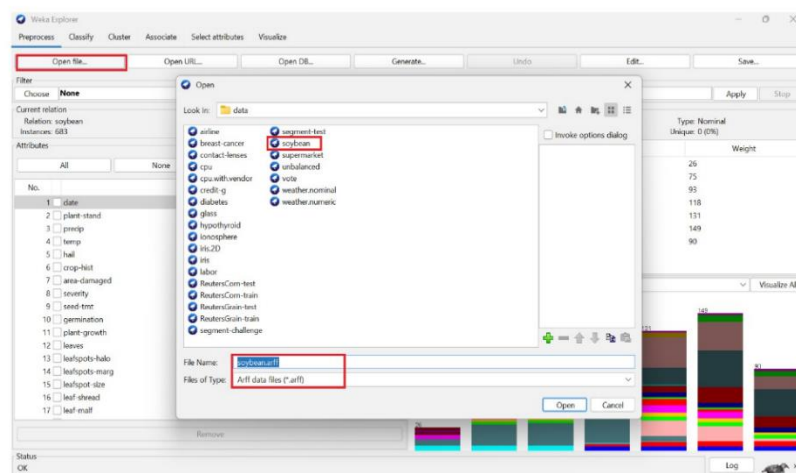
Procedure:

Step 1: Start WEKA

- Open the WEKA tool.
- Click on Explorer.

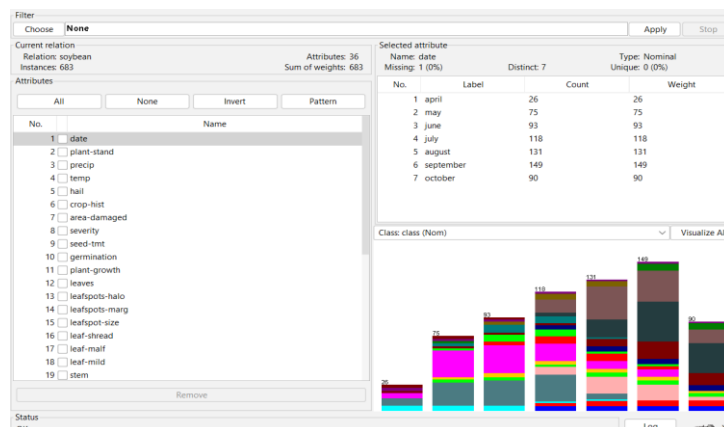
Step 2: Load the Dataset

- Go to the Preprocess tab.
- Click Open File.
- Select the dataset soybean.arff.
- Click Open.



Step 3: Verify Dataset

- Check that all instances and attributes are loaded correctly.
- Observe dataset information in the Preprocess panel.



Date:

Roll No.:

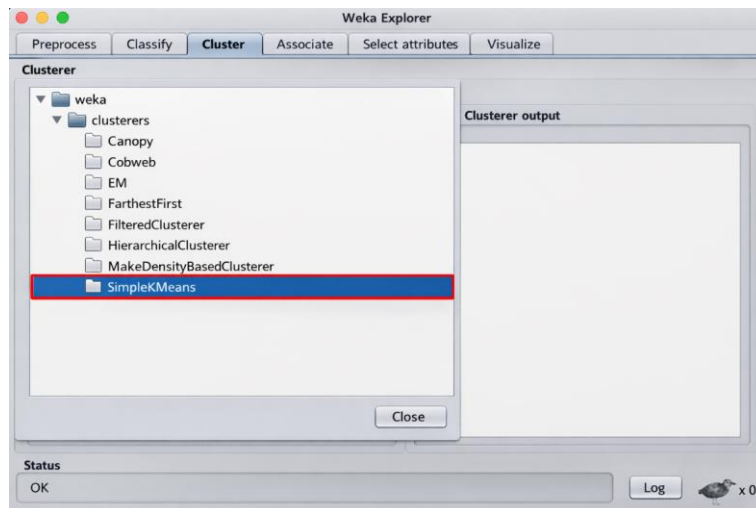
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Step 4: Move to Cluster Tab

- Click on the Cluster tab.

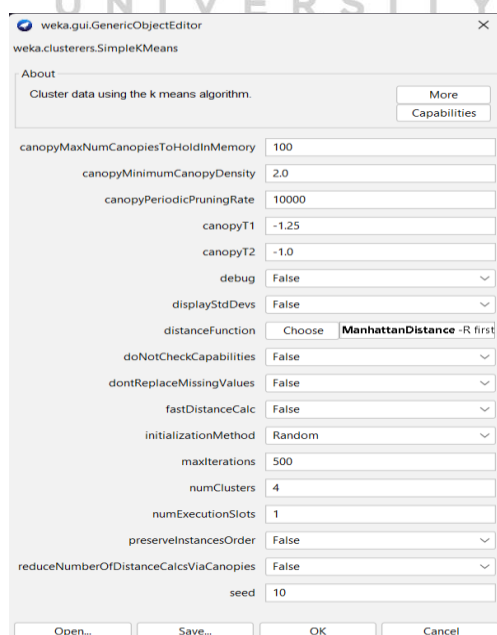
Step 5: Choose Clustering Algorithm

- Click Choose.
- Select SimpleKMeans.



Step 6: Set Algorithm Parameters

- Set Number of Clusters (K) = 4.
- Set Distance Function = Manhattan Distance.



Date:

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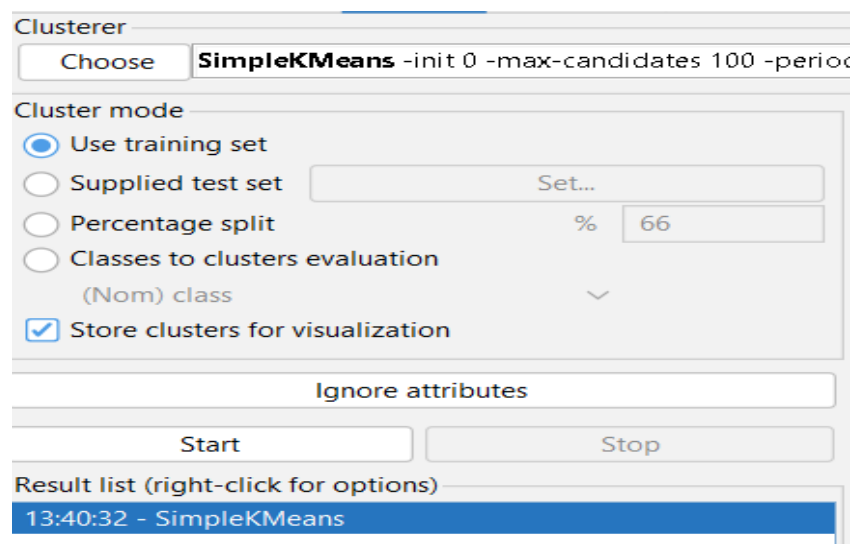
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Step 7: Set Cluster Mode

- Under Cluster mode, select Use training set.
- From the available options, select Use training set.
- This option tells WEKA to perform clustering using the same dataset that was loaded earlier.

Step 8: Run the Algorithm

- Click Start.
- Wait for WEKA to process the dataset.



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Step 9: Viewing the Results

After execution, WEKA displays the clustering results including:

- Clustering model

```
=== Clustering model (full training set) ===

kMeans
=====

Number of iterations: 9
Sum of within cluster distances: 5693.0

Initial starting points (random):

Cluster 0: september,normal,gt-norm,norm,yes,same-1st-two-yrs,whole-field,severe,fungicide,80-89,norm,abnorm,no-yellow-halos,w-s-marg,gt-1/8,absent,absent,absent
Cluster 1: june,lt-normal,gt-norm,norm,yes,same-1st-sev-yrs,whole-field,pot-severe,none,lt-80,norm,abnorm,no-yellow-halos,w-s-marg,gt-1/8,present,absent,absent,i
Cluster 2: october,normal,gt-norm,lt-norm,yes,same-1st-two-yrs,upper-areas,minor,fungicide,80-89,norm,abnorm,no-yellow-halos,w-s-marg,lt-1/8,absent,absent,absent
Cluster 3: september,lt-normal,gt-norm,gt-norm,yes,same-1st-sev-yrs,low-areas,minor,fungicide,lt-80,norm,abnorm,no-yellow-halos,w-s-marg,gt-1/8,absent,absent,ab

Missing values globally replaced with mean/mode
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- Cluster centroids

Final cluster centroids:					
Attribute	Full Data (683.0)	Cluster# 0 (237.0)	1 (281.0)	2 (41.0)	3 (124.0)
date	september	september	august	october	september
plant-stand	normal	lt-normal	normal	normal	lt-normal
precip	gt-norm	gt-norm	gt-norm	gt-norm	gt-norm
temp	norm	norm	norm	gt-norm	gt-norm
hail	yes	yes	yes	yes	yes
crop-hist	same-1st-two-yrs	same-1st-two-yrs	same-1st-two-yrs	same-1st-sev-yrs	same-1st-sev-yrs
area-damaged	low-areas	low-areas	whole-field	whole-field	whole-field
severity	pot-severe	pot-severe	pot-severe	minor	minor
seed-tmt	none	none	none	none	fungicide
germination	80-89	80-89	80-89	80-89	lt-80
plant-growth	norm	abnorm	norm	norm	norm
leaves	abnorm	abnorm	abnorm	norm	abnorm
leafspots-halo	no-yellow-halos	absent	no-yellow-halos	no-yellow-halos	no-yellow-halos
leafspots-marg	w-s-marg	dna	w-s-marg	w-s-marg	w-s-marg
leafspots-size	gt-1/8	dna	gt-1/8	gt-1/8	gt-1/8
leaf-shread	absent	absent	absent	absent	absent
leaf-malf	absent	absent	absent	absent	absent
leaf-mild	absent	absent	absent	absent	absent
stem	abnorm	abnorm	norm	abnorm	norm
lodging	yes	yes	yes	yes	yes
stem-cankers	absent	above-sec-nde	absent	absent	absent
canker-lesion	dna	dk-brown-blk	dna	tan	dna
fruiting-bodies	absent	absent	absent	absent	absent
external-decay	absent	absent	absent	absent	absent
mycelium	absent	absent	absent	absent	absent
int-discolor	none	none	none	none	none
sclerotia	absent	absent	absent	absent	absent
fruit-pods	norm	norm	norm	diseased	norm
fruit-spots	absent	absent	absent	brown-w/blk-specks	absent
seed	norm	norm	norm	abnorm	norm
mold-growth	absent	absent	absent	absent	absent
seed-discolor	absent	absent	absent	present	absent
seed-size	norm	norm	norm	norm	norm
shriveling	absent	absent	absent	absent	absent
roots	norm	norm	norm	norm	norm
class	brown-spot	phytophthora-rot	brown-spot	purple-seed-stain	alternarialeaf-spot

- Number of instances and cluster distribution

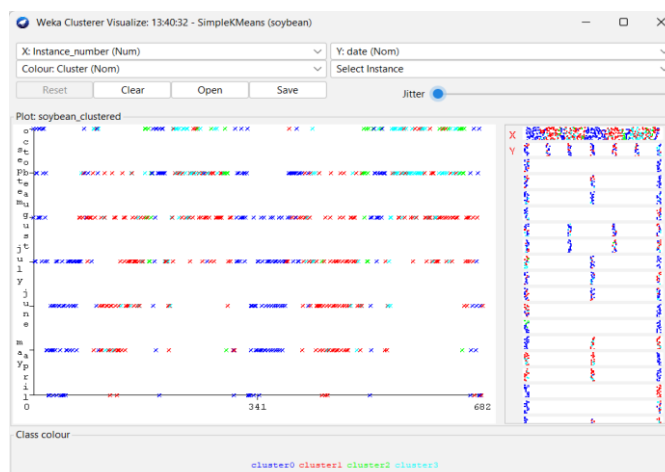
```
=== Model and evaluation on training set ===

Clustered Instances

0      237 ( 35%)
1      281 ( 41%)
2       41 (  6%)
3      124 ( 18%)
```

Step 10: Visualizing Cluster

- Use the Visualize Cluster Assignments option to view the graphical representation of clusters.
- Observe how the dataset instances are grouped into four clusters based on Manhattan distance.



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Conclusion:

The clustering experiment was successfully performed using the SimpleKMeans algorithm in WEKA. The soybean dataset, which contains information about different soybean plant conditions and diseases, was used for this experiment. The algorithm grouped the 683 instances into 4 clusters using the Manhattan distance measure based on the similarity among attribute values. The clustering results showed the cluster centroids, number of instances in each cluster, and cluster distribution, which helped in understanding how the data was organized into meaningful groups. Thus, the experiment demonstrates that K-Means clustering is an effective unsupervised learning technique for identifying patterns and grouping similar data instances in a dataset.

CO Mapped: CO5

POs Mapped: PO1, PO2, PO3, PO5, PO8, PO9, PO11

PSOs Mapped: PSO2



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Practice No.: 10

Apply Hierarchical clustering process on dataset.

Aim: To apply the Hierarchical clustering algorithm on a dataset and analyze how the data instances are grouped based on similarity using WEKA.

Tool Used:

Software: Weka (Waikato Environment for Knowledge Analysis).

Operating System: Windows / Linux

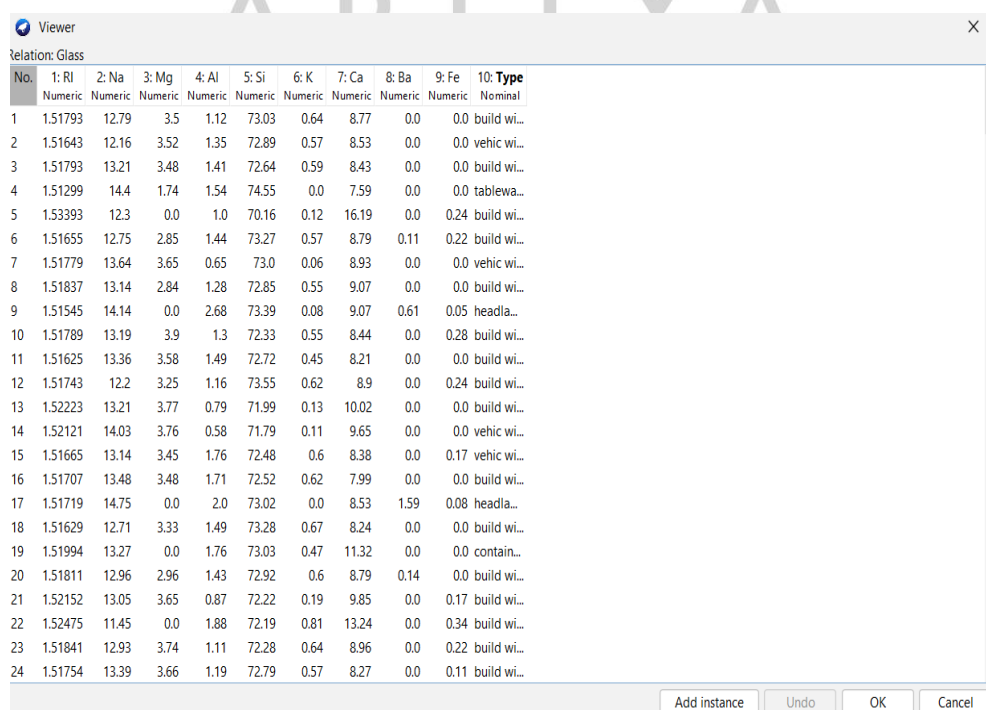
Algorithm Used: K-Means Clustering (SimpleKMeans in WEKA)

Dataset Used:

The dataset used is glass.arff from the WEKA data repository.

The Glass Dataset contains chemical composition data of different types of glass. Each instance represents a glass sample with attributes such as refractive index and concentrations of elements like sodium, magnesium, aluminum, silicon, potassium, calcium, barium, and iron.

The dataset contains 214 instances and 10 attributes, making it suitable for clustering experiments. It is commonly used to analyze and group glass samples based on their chemical properties.



No.	1: RI	2: Na	3: Mg	4: Al	5: Si	6: K	7: Ca	8: Ba	9: Fe	10: Type
	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Nominal
1	1.51793	12.79	3.5	1.12	73.03	0.64	8.77	0.0	0.0	build wi...
2	1.51643	12.16	3.52	1.35	72.89	0.57	8.53	0.0	0.0	vehic wi...
3	1.51793	13.21	3.48	1.41	72.64	0.59	8.43	0.0	0.0	build wi...
4	1.51299	14.4	1.74	1.54	74.55	0.0	7.59	0.0	0.0	tablewa...
5	1.53393	12.3	0.0	1.0	70.16	0.12	16.19	0.0	0.24	build wi...
6	1.51655	12.75	2.85	1.44	73.27	0.57	8.79	0.11	0.22	build wi...
7	1.51779	13.64	3.65	0.65	73.0	0.06	8.93	0.0	0.0	vehic wi...
8	1.51837	13.14	2.84	1.28	72.85	0.55	9.07	0.0	0.0	build wi...
9	1.51545	14.14	0.0	2.68	73.39	0.08	9.07	0.61	0.05	headla...
10	1.51789	13.19	3.9	1.3	72.33	0.55	8.44	0.0	0.28	build wi...
11	1.51625	13.36	3.58	1.49	72.72	0.45	8.21	0.0	0.0	build wi...
12	1.51743	12.2	3.25	1.16	73.55	0.62	8.9	0.0	0.24	build wi...
13	1.52223	13.21	3.77	0.79	71.99	0.13	10.02	0.0	0.0	build wi...
14	1.52121	14.03	3.76	0.58	71.79	0.11	9.65	0.0	0.0	vehic wi...
15	1.51665	13.14	3.45	1.76	72.48	0.6	8.38	0.0	0.17	vehic wi...
16	1.51707	13.48	3.48	1.71	72.52	0.62	7.99	0.0	0.0	build wi...
17	1.51719	14.75	0.0	2.0	73.02	0.0	8.53	1.59	0.08	headla...
18	1.51629	12.71	3.33	1.49	73.28	0.67	8.24	0.0	0.0	build wi...
19	1.51994	13.27	0.0	1.76	73.03	0.47	11.32	0.0	0.0	contain...
20	1.51811	12.96	2.96	1.43	72.92	0.6	8.79	0.14	0.0	build wi...
21	1.52152	13.05	3.65	0.87	72.22	0.19	9.85	0.0	0.17	build wi...
22	1.52475	11.45	0.0	1.88	72.19	0.81	13.24	0.0	0.34	build wi...
23	1.51841	12.93	3.74	1.11	72.28	0.64	8.96	0.0	0.22	build wi...
24	1.51754	13.39	3.66	1.19	72.79	0.57	8.27	0.0	0.11	build wi...

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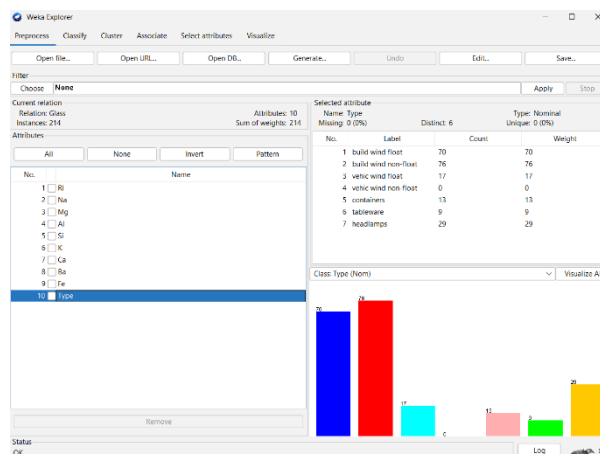
Step 1: Start WEKA

- Open WEKA Tool
- Click on Explore

- Go to the Preprocess tab.
- Click Open File.
- Select the dataset glass.arff.
- Click Open.



- Check that all instances and attributes are loaded correctly.
- Observe the dataset information displayed in the Preprocess panel.



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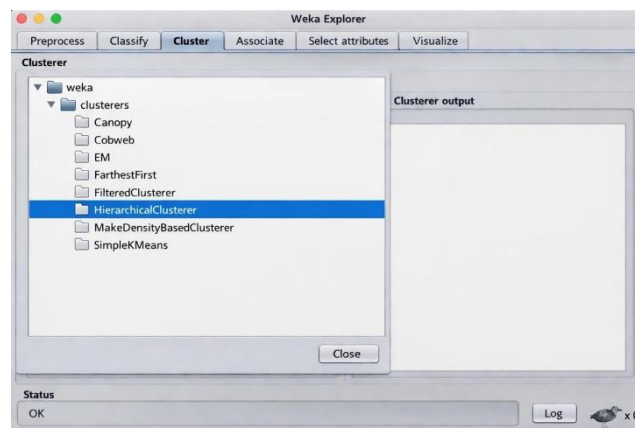
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Step 4: Move to Cluster Tab

- Click on the Cluster tab.

Step 5: Choose Clustering Algorithm

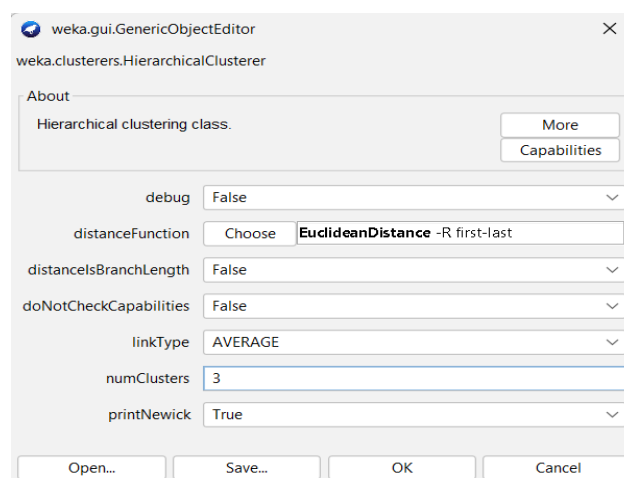
- Click Choose.
- Select HierarchicalClusterer.



Step 6: Set Algorithm Parameters

Open the HierarchicalClusterer settings and configure:

- Number of Clusters = 3 (or any suitable number)
- Link Type = Average



Date:

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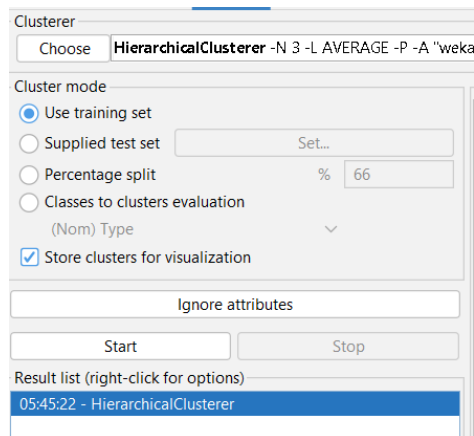
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Step 7: Set Cluster Mode

- Under Cluster mode, select Use training set.
- This option allows WEKA to perform clustering using the loaded dataset.

Step 8: Run the Algorithm

- Click Start.
- Wait for WEKA to process the dataset.



Step 9: Observe Output

WEKA displays the clustering results including:

- Hierarchical cluster tree structure
- Cluster assignments
- Cluster distribution



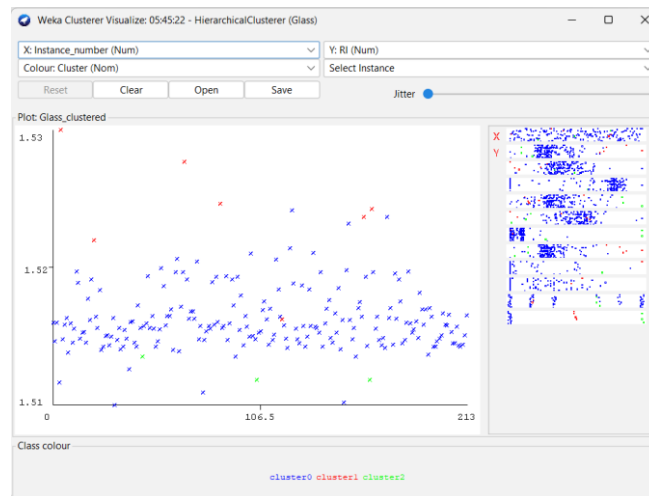
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Step 10: Analyze the Clusters

- Analyze how the glass samples are grouped into clusters based on similarities in their chemical composition.



Conclusion :

The hierarchical clustering experiment was successfully performed using the HierarchicalClusterer algorithm in WEKA. The Glass Dataset was used for this experiment, which contains information about the chemical composition of different glass samples. The algorithm grouped similar glass samples based on their attribute similarities and formed clusters in a hierarchical tree structure. This method helped in understanding the relationships among the data instances by progressively merging similar clusters. Thus, the experiment demonstrates that Hierarchical Clustering is an effective unsupervised learning technique for discovering patterns and grouping similar data points in a dataset.

CO Mapped: CO5

POs Mapped: PO1, PO2, PO3, PO5, PO8, PO9, PO11

PSOs Mapped: PSO2